

COMP 4423: Machine Learning and Deep Learning

Easy Computer Vision

Xiaoyong Wei (魏驍勇)
x1wei@polyu.edu.hk

The Hong Kong Polytechnic University
香港理工大學
Opening Minds • Shaping the Future
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Outline

- Traditional machine learning vs. deep learning
- Gradient decent
- Neural networks
- Deep neural networks
- Convolutional neural networks (CNN)
- Layers, pooling, and activation
- AlexNet, VGG, and ResNet



Traditional classification methods work well for simple tasks. Models are usually built in a controlled environment (e.g., lab setting) to eliminate the variations of illumination, viewpoints, scales, and so on.

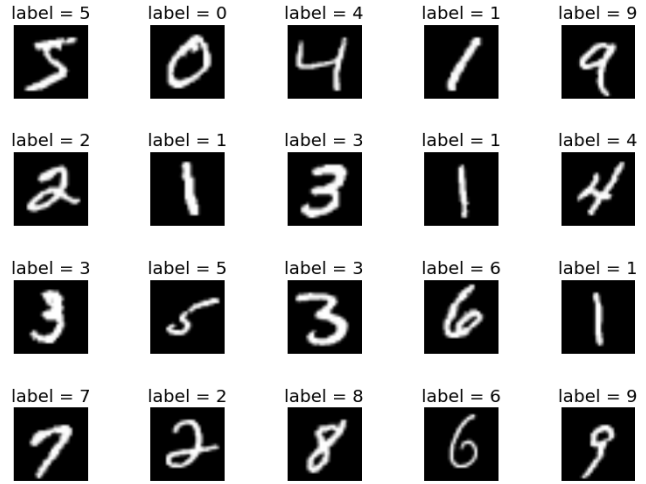
Popular traditional datasets



Olivetti Face Dataset, AT&T

Popular traditional datasets

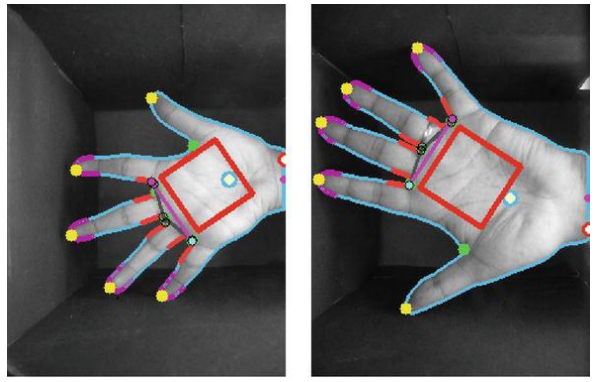




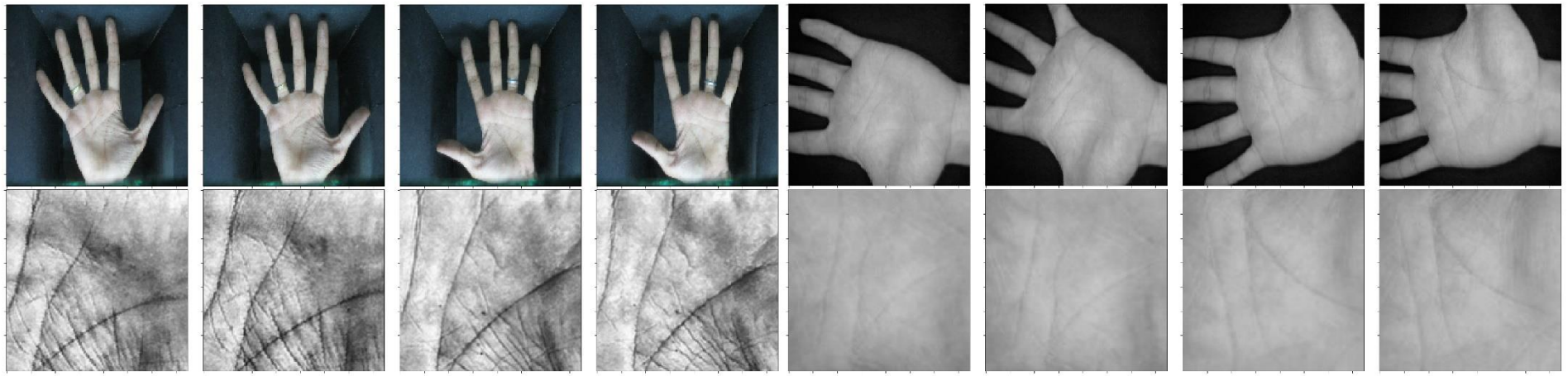
MINIST Handwritten Digits

Popular traditional datasets







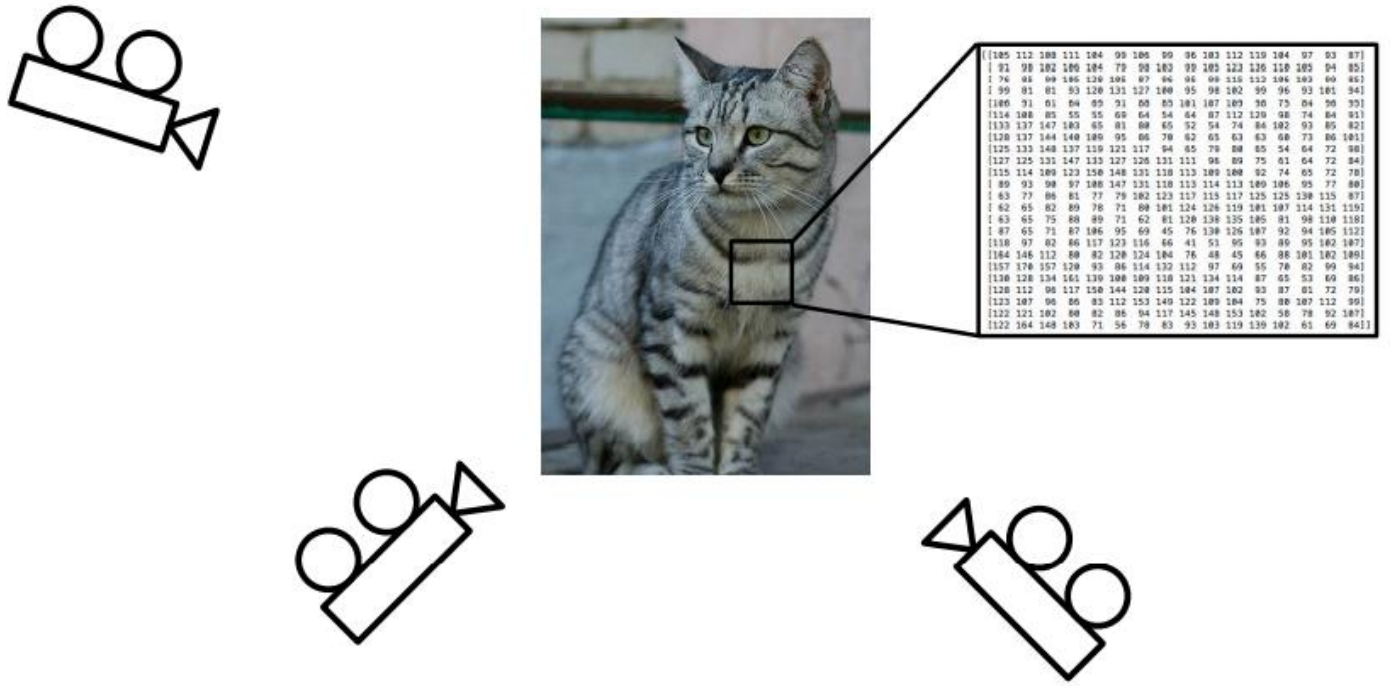


Palmprint Acquisition and Datasets



However, in real applications, those are inevitable.

Viewpoints



All pixels change when the camera moves!

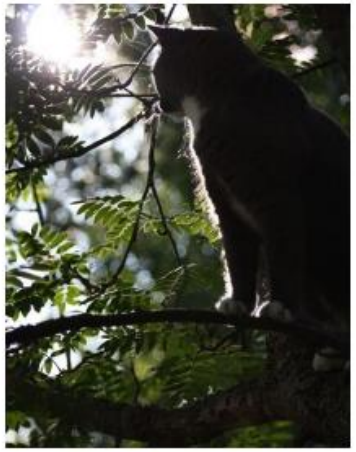
Fei-Fei Li, Ranjay Krishna, Danfei Xu, Image Classification: A Core Task in Computer Vision

Illumination









Fei-Fei Li, Ranjay Krishna, Danfei Xu, Image Classification: A Core Task in Computer Vision

Occlusions



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Background Clutter



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Intra-class Variations



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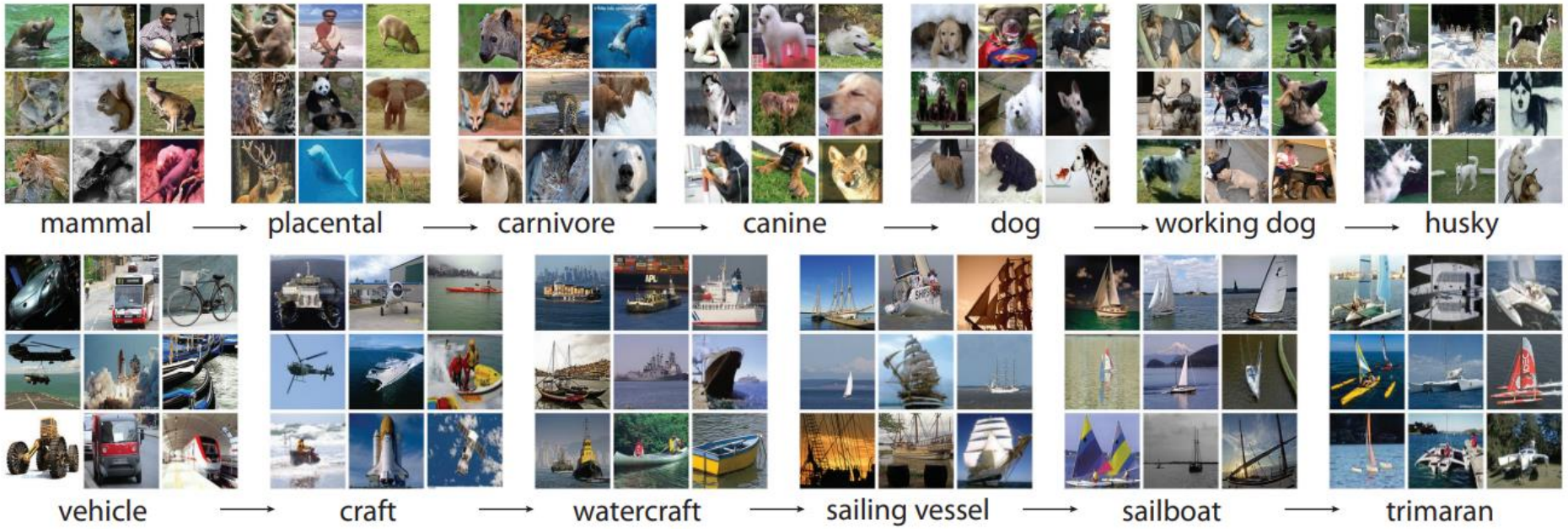
Hand Gesture Recognition





ImageNet

ImageNet: 12 subtrees with 5247 synsets and 3.2 million images in total



J. Deng, W. Dong, R. Socher, L. Li, Kai Li and Li Fei-Fei, "ImageNet: A large-scale hierarchical image database," 2009 IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248-255, doi: 10.1109/CVPR.2009.5206848.

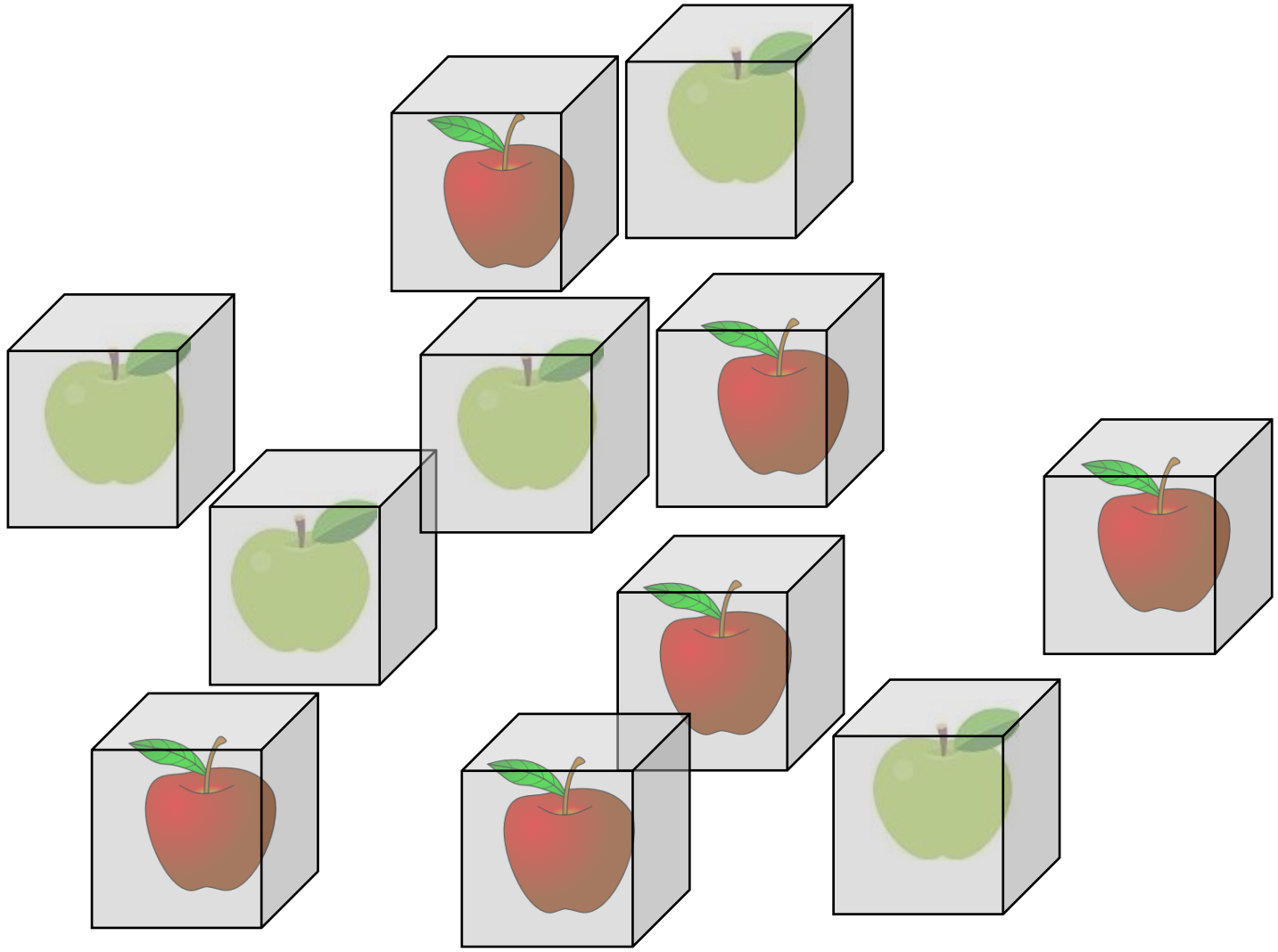


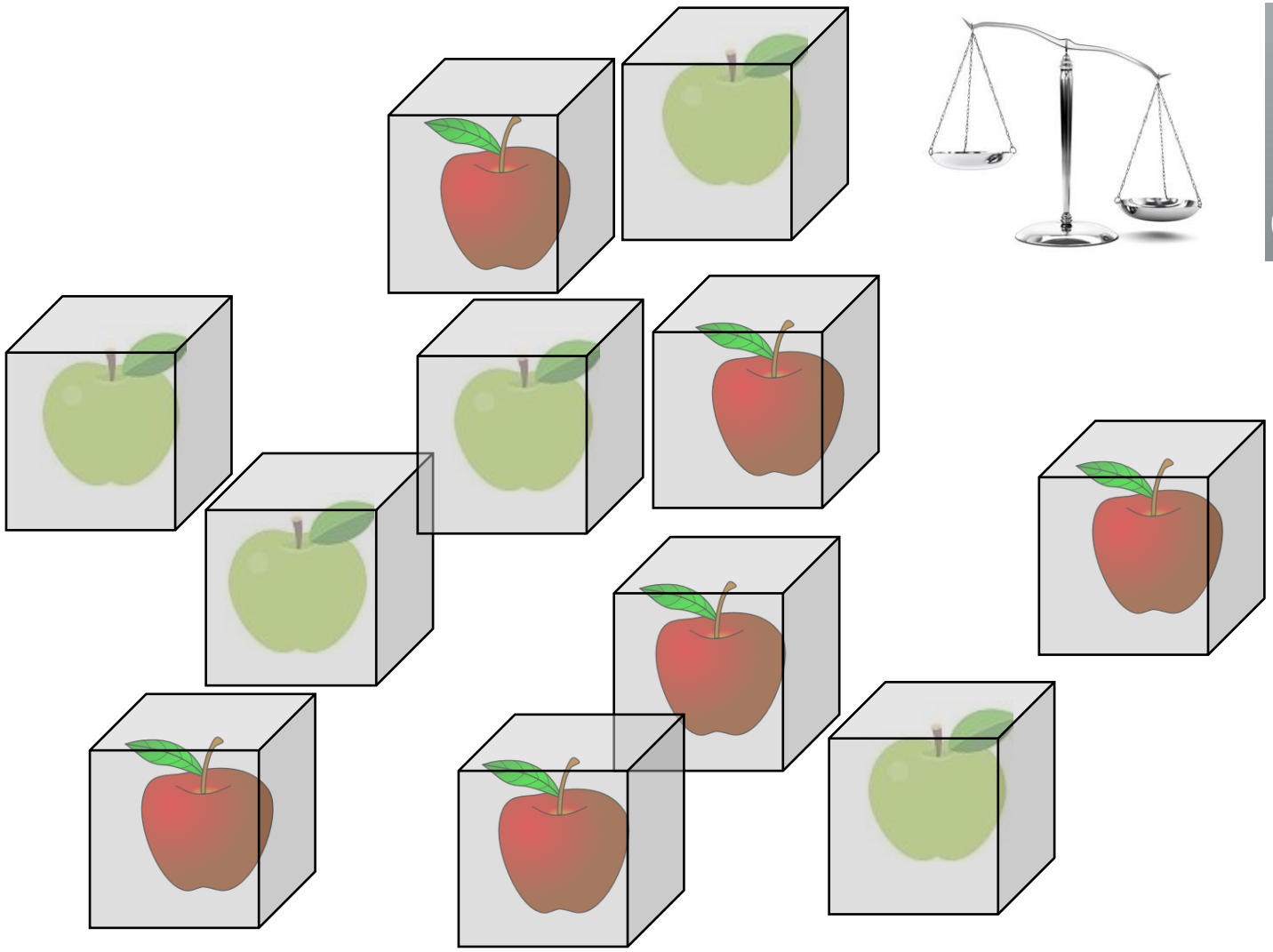
Deep Learning is a popular solution to address these challenges.

(This is what you're waiting for. LOL!)



Let’s start by reviewing the learning of decision boundary through an example – to classify the red and green apples.



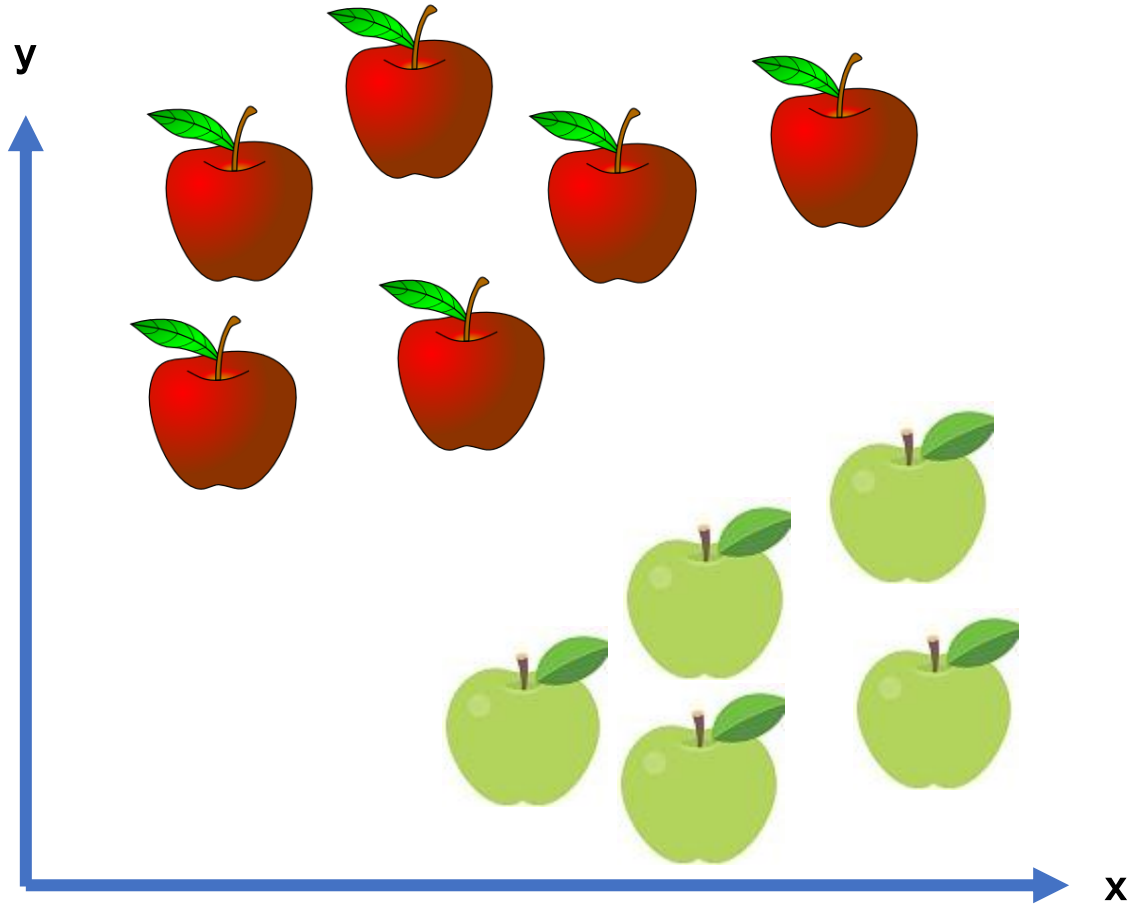




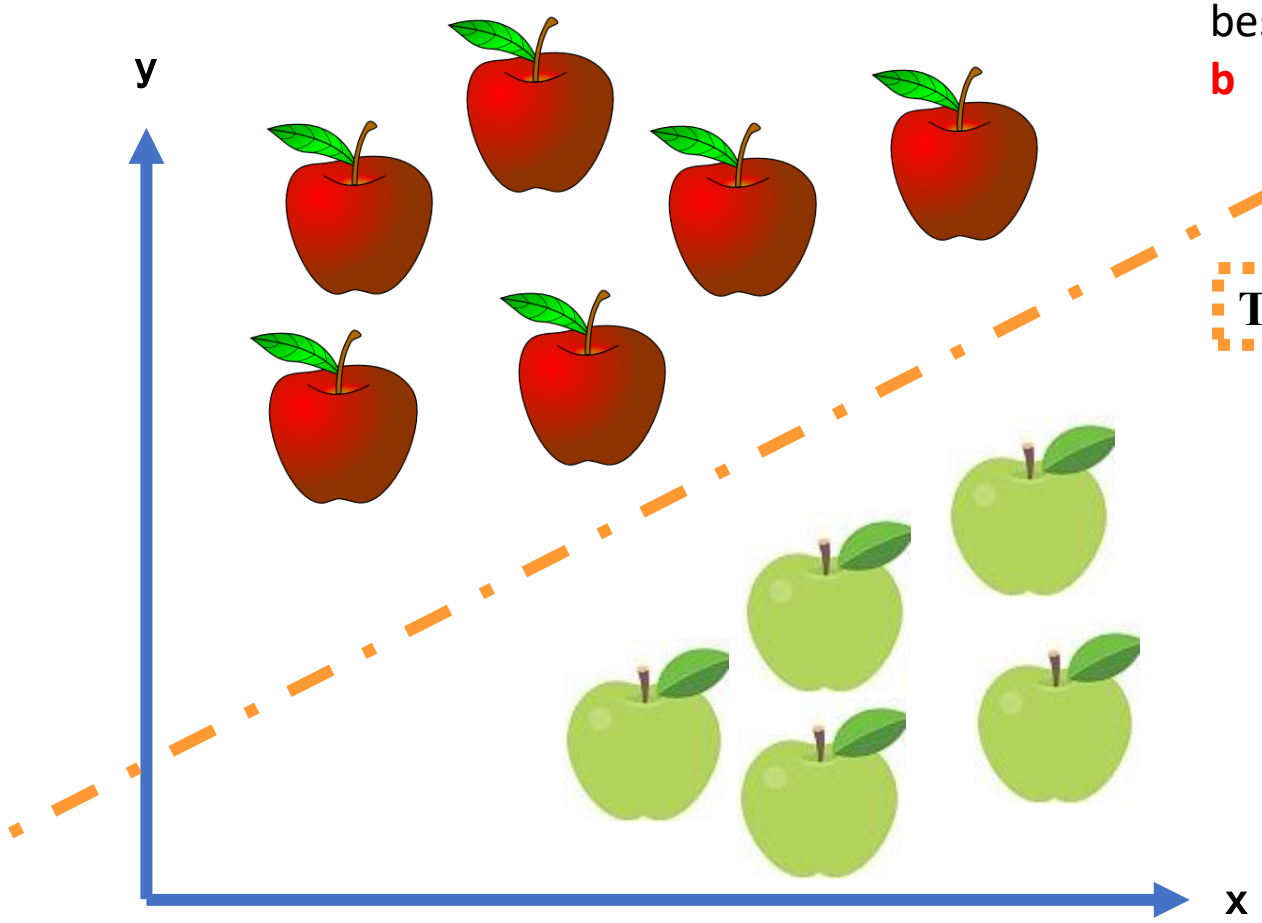
List of apples

No.	x	y	others	Color (z)
1	9	53	…	Red (1)
2	25	45	…	Green (-1)
3	225	56.7	…	Red (1)
4	576	52.9	…	Green (-1)
5	676	60.2	…	Red (1)
6	900	55.7	…	Green (-1)
7	…	…	…	…

Apple Space



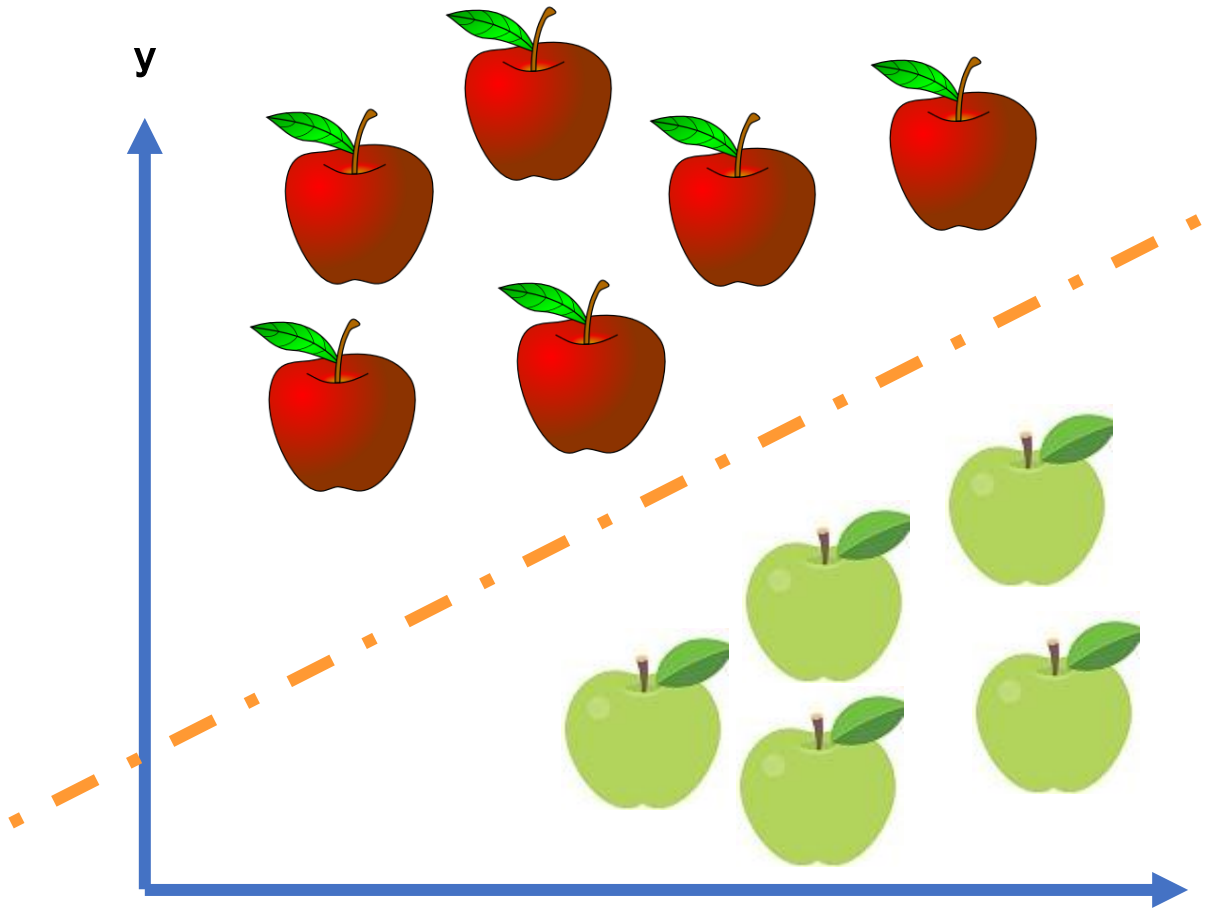
Apple Space



To find the best line dividing the two groups of apples is to find the best **parameters** of **a** and **b**

The line: y =a*x+b

Apple Space



To find the best line dividing the two groups of apples is to find the best **parameters** of **a** and **b**

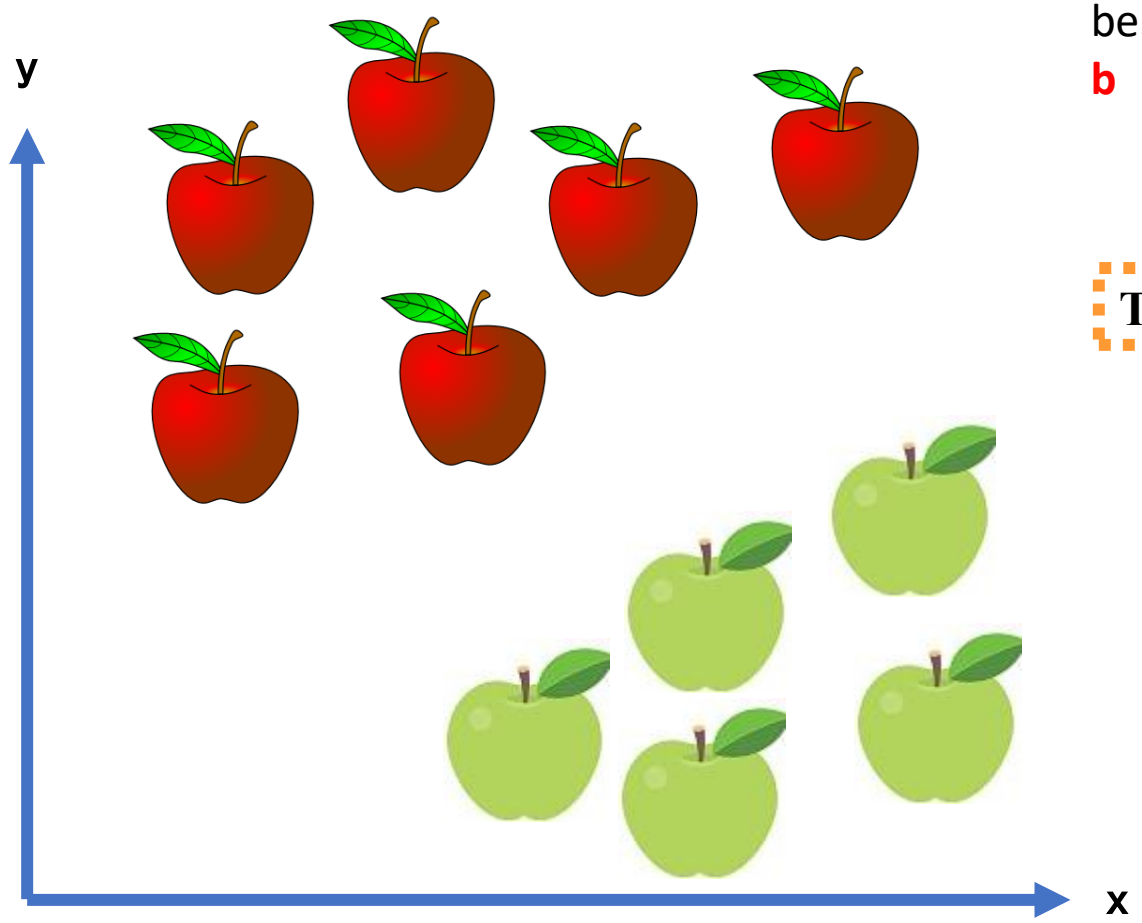
The line: y =a*x+b

The model:

z=a*x-y+b

Outputs 1 if z>0
Outputs -1 if z<=0

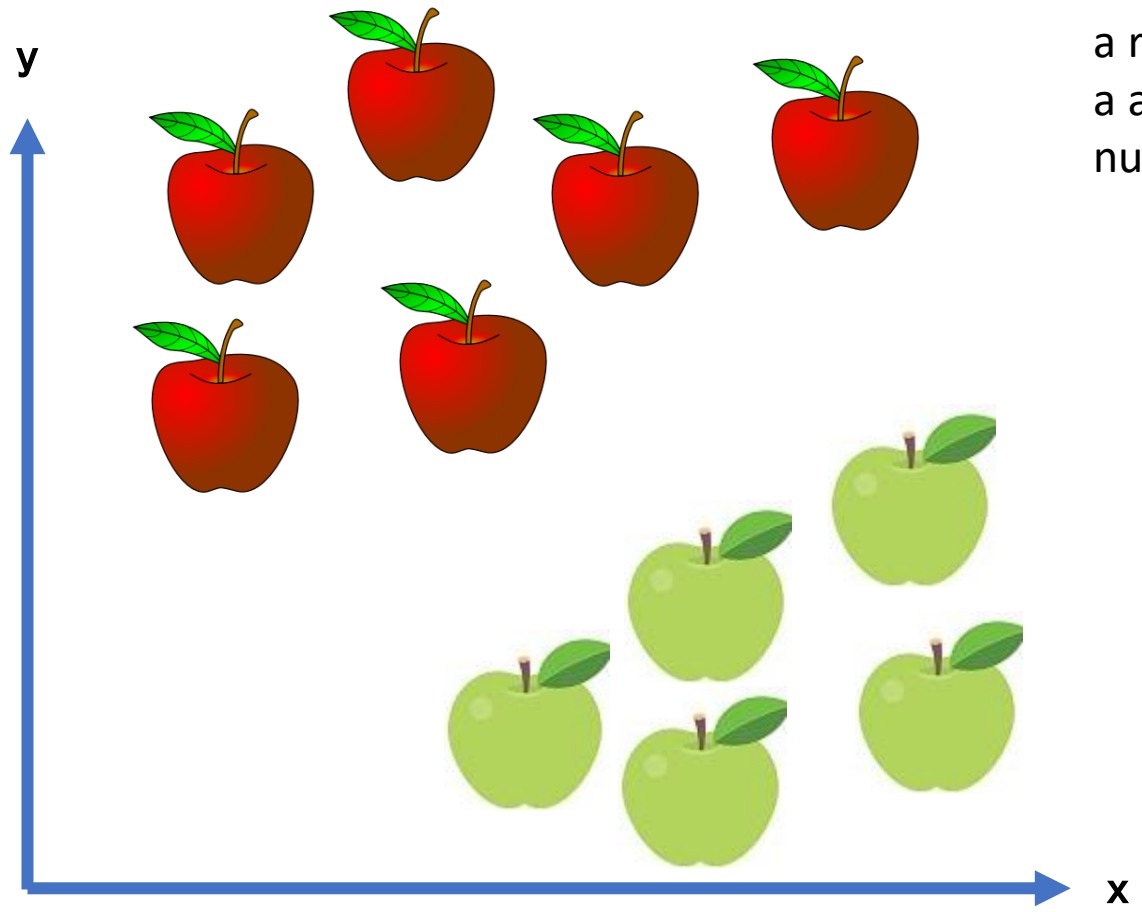
Apple Space



To find the best line dividing the two groups of apples is to find the best **parameters** of **a** and **b**

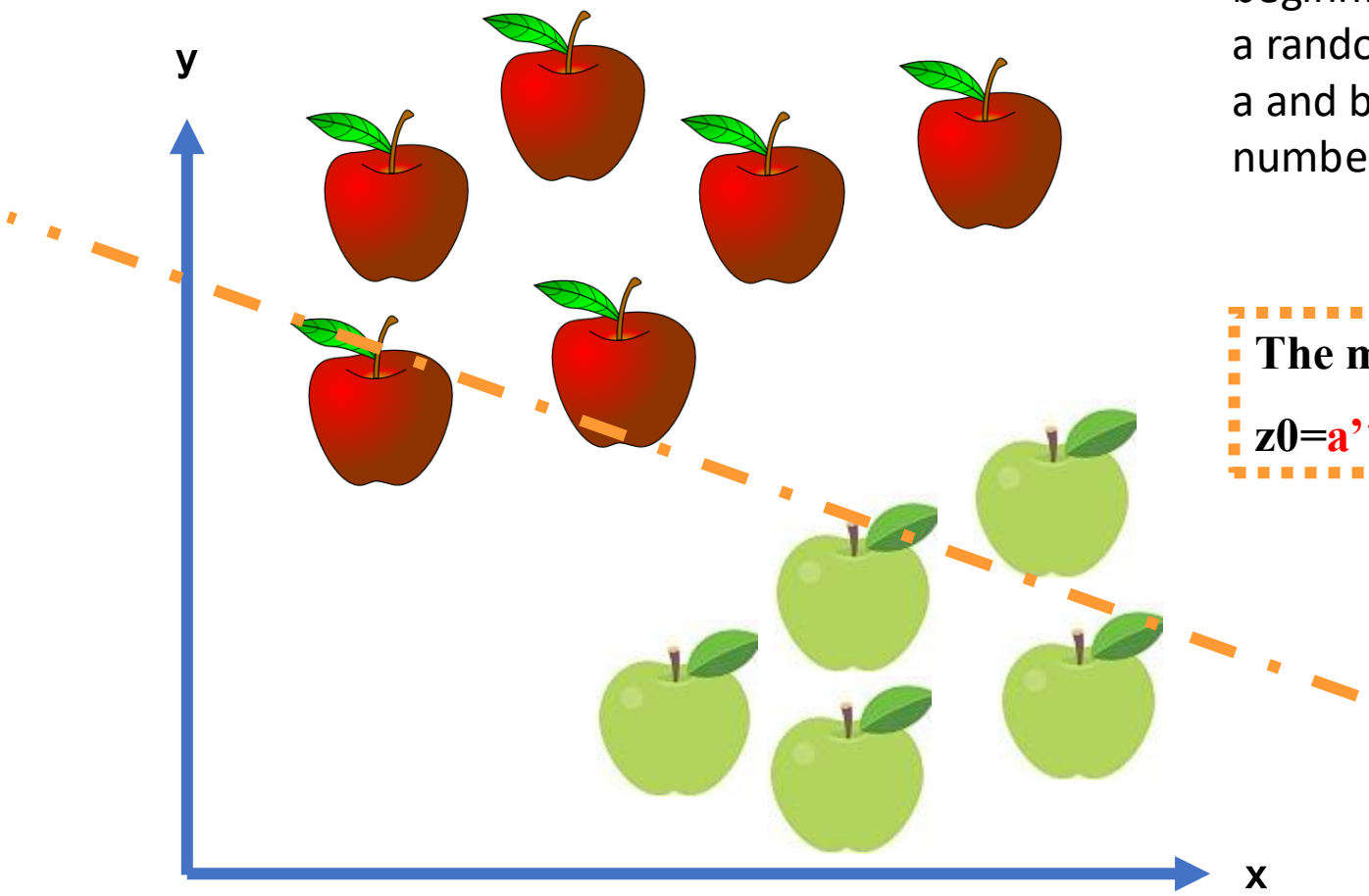
The line: y =a*x+b

Apple Space



Initialization
Without knowing which line is the best at the beginning, we can pick a random one by setting a and be with random numbers a' and b'.

Apple Space



Initialization

Without knowing which line is the best at the beginning, we can pick a random one by setting a and be with random numbers a’ and b’.

The model:

z0=a’*x-y+b’



How can we evaluate how good the **model** (a' and b') is?

Intuitively, we can compare the **prediction** z' to the **ground truth label** z using $(z'-z)^2$. By applying to all N samples, we have a **loss function**

$$L(a', b') = \frac{1}{N}\sum_{i=1}^{N}(z'_i - z_i)^2$$



With the “goodness” evaluated, we can update a’ and b’ by replacing them with better ones.

The updating process is so called **learning**.



But, how?



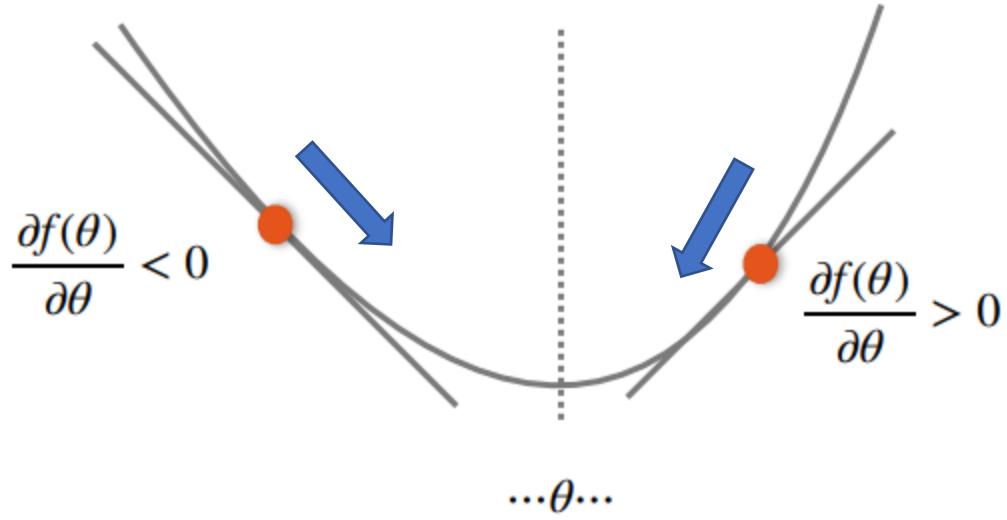
The best parameters are the ones that minimize the loss function L. The optimal parameters can thus be found at where the **gradients** of L are zeros

$$\frac{\partial L}{\partial a} = 0, \qquad \frac{\partial L}{\partial b} = 0.$$

We can update a’ and b’ by pushing the gradients towards zeros!

$$a' = a' - \frac{\partial L}{\partial a'}$$

$$b' = b' - \frac{\partial L}{\partial b'}$$



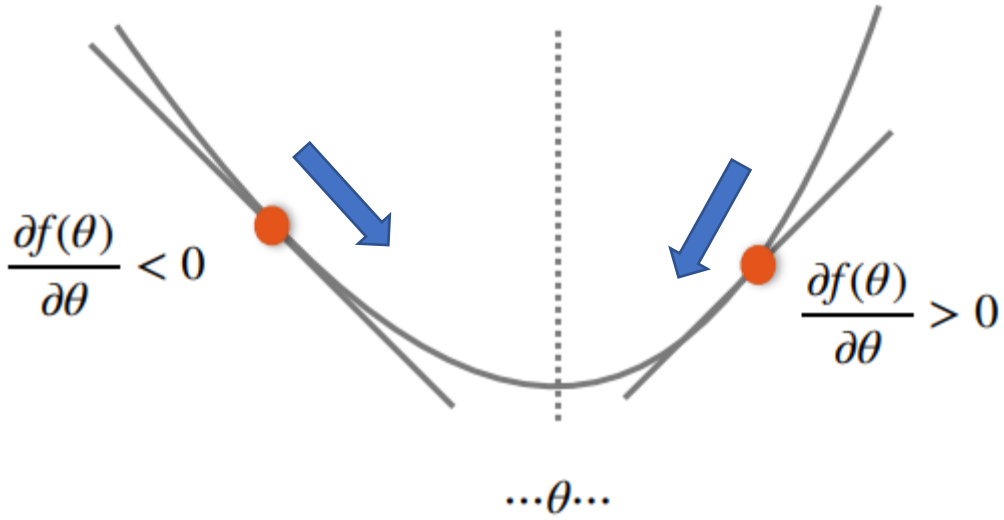
Gradient Decent

We can update a' and b' by pushing the gradients towards zeros!

$$a' = a' - \delta \frac{\partial L}{\partial a'}$$

$$b' = b' - \delta \frac{\partial L}{\partial b'}$$

Learning Rate



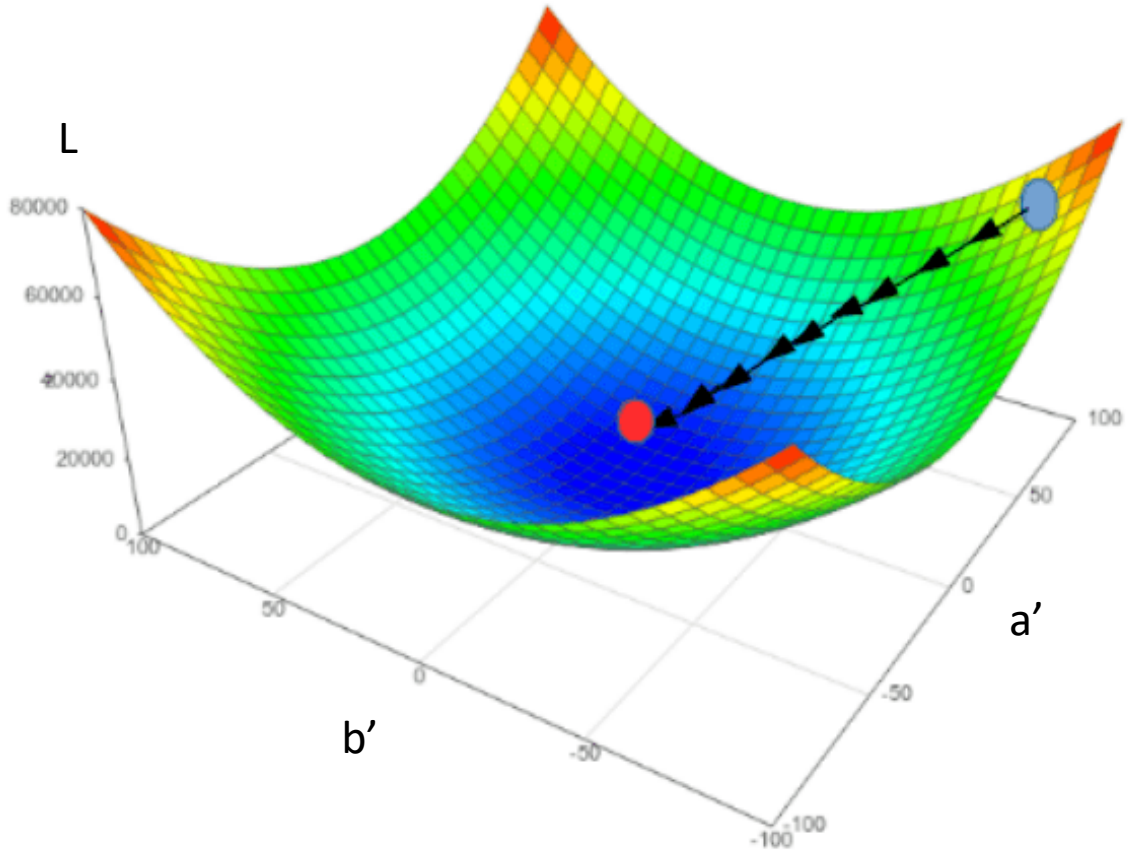
Gradient Decent

We can update a' and b' by pushing the gradients towards zeros!

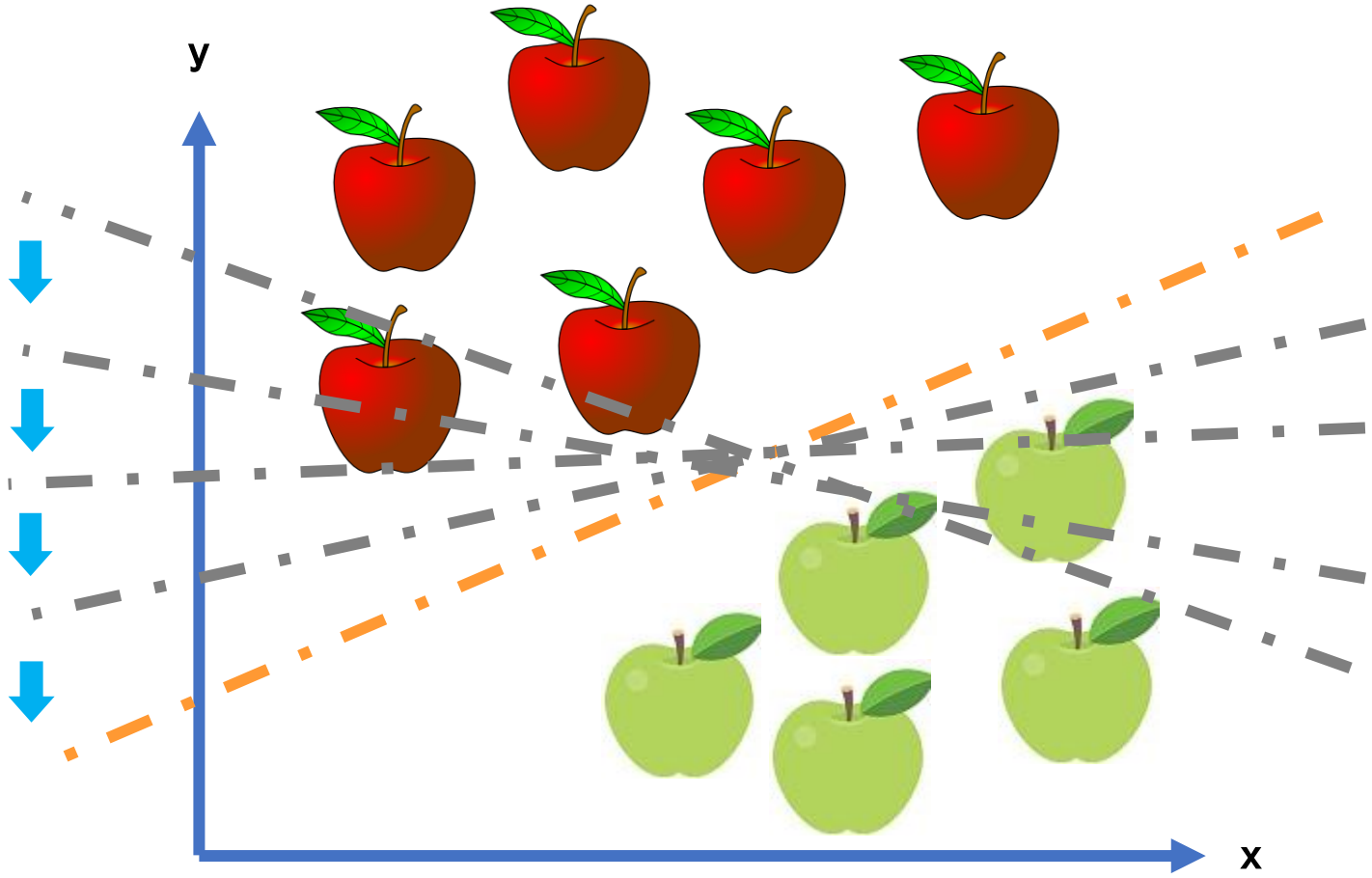
$$a' = a' - \delta \frac{\partial L}{\partial a'}$$

$$b' = b' - \delta \frac{\partial L}{\partial b'}$$

Learning Rate



Apple Space



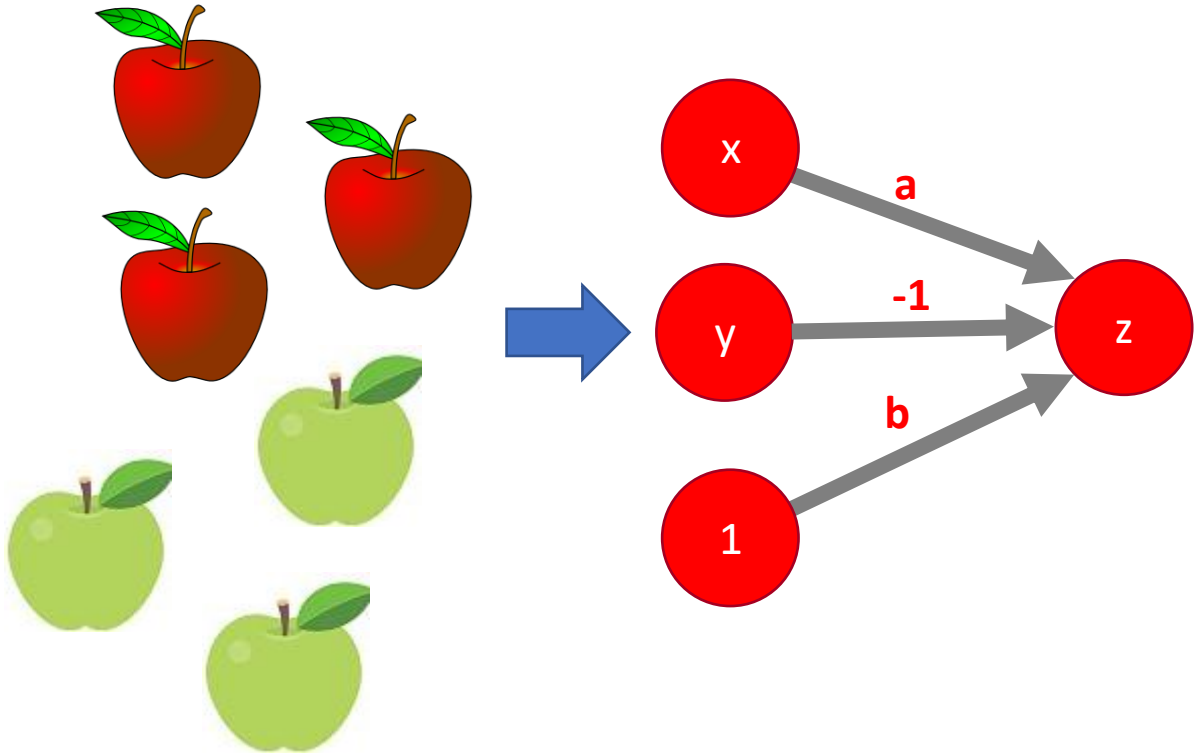


Machine learning is a process to find the best set of **parameters** that fits into a **model/hypothesis**.
The learning is usually conducted by updating the initial parameters with a learning rate towards the optimal of a **loss function**. **Gradient Decent** is one of the most popular updating strategies.

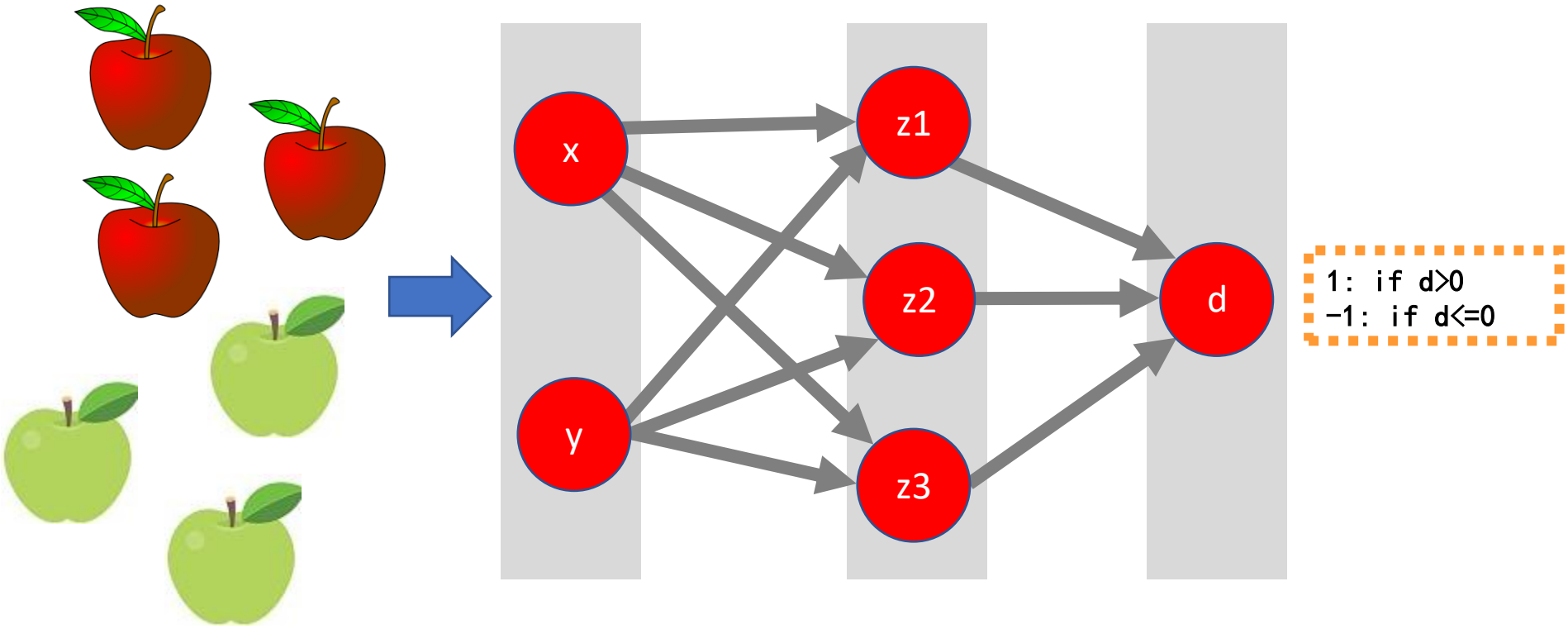


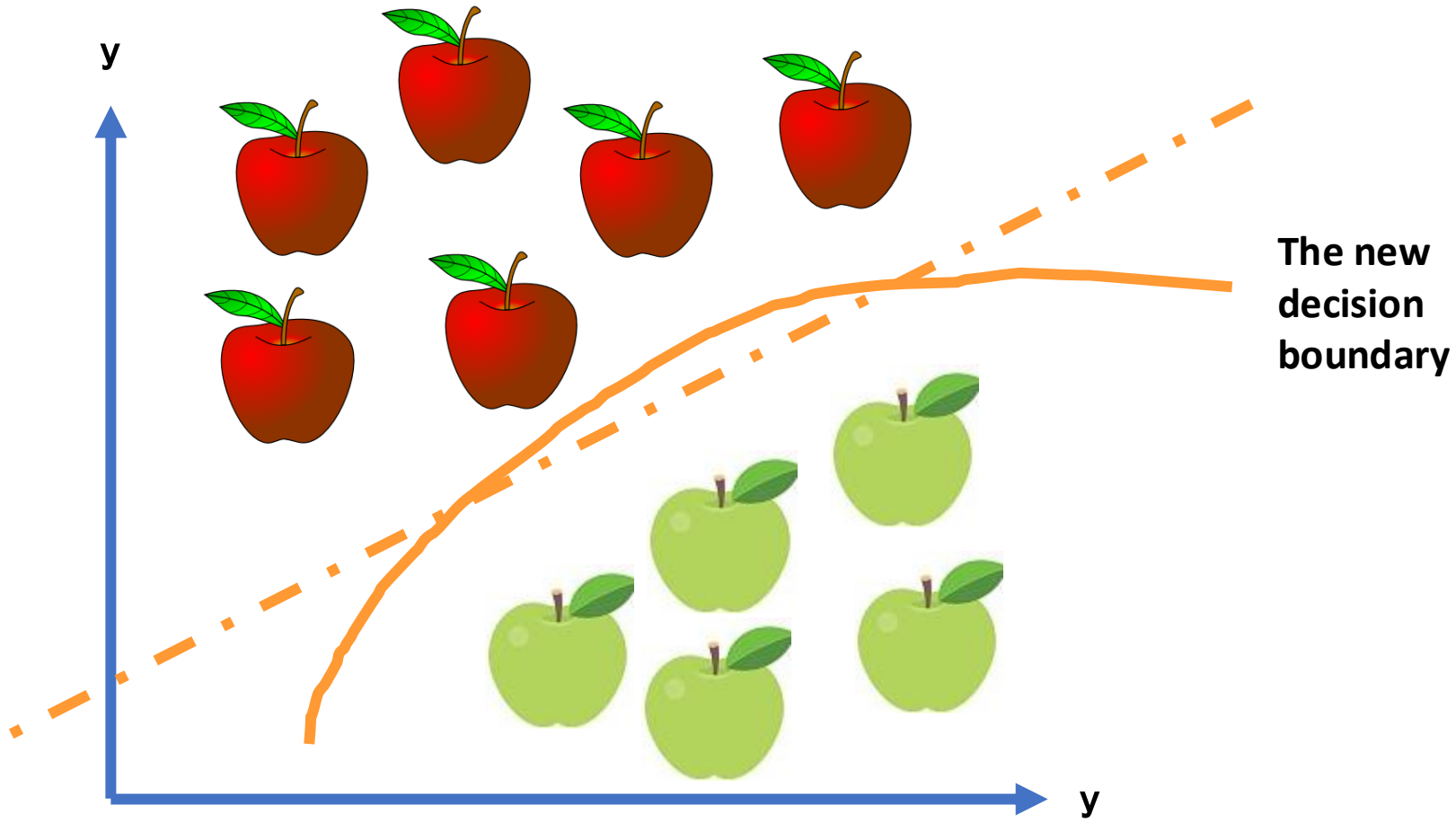
Let’s implement the learning using **neural networks**.

Neural Network Version of the Model



Neural Network Version of the Model

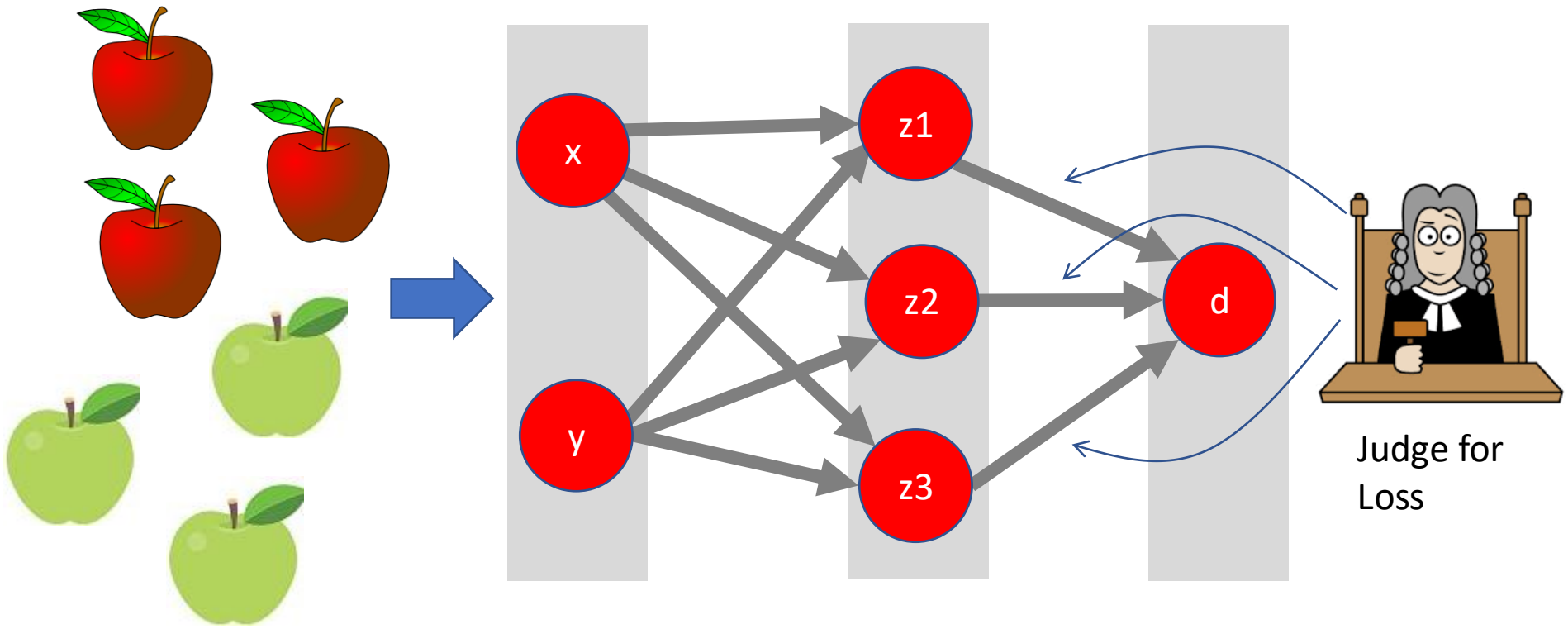



y
y
The new
decision
boundary

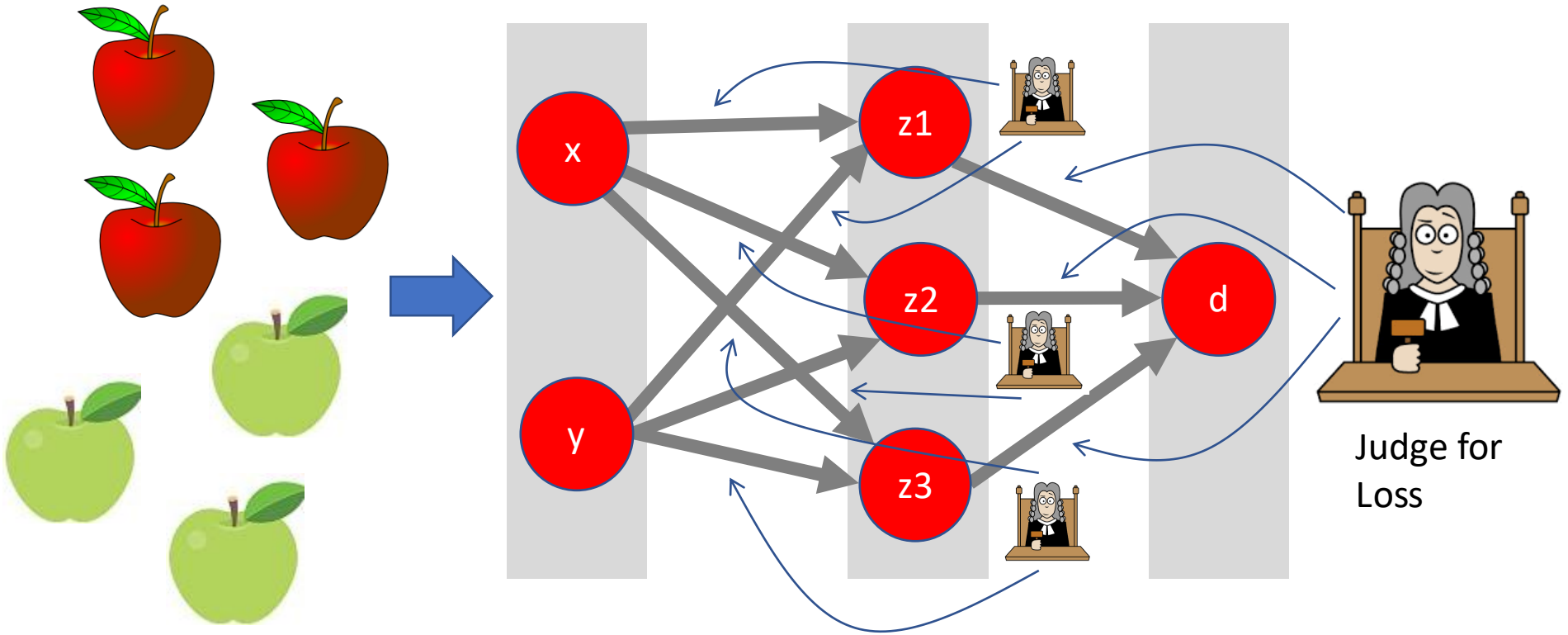


How is the learning conducted with more layers and weights?

Gradient Decent on Neural Networks



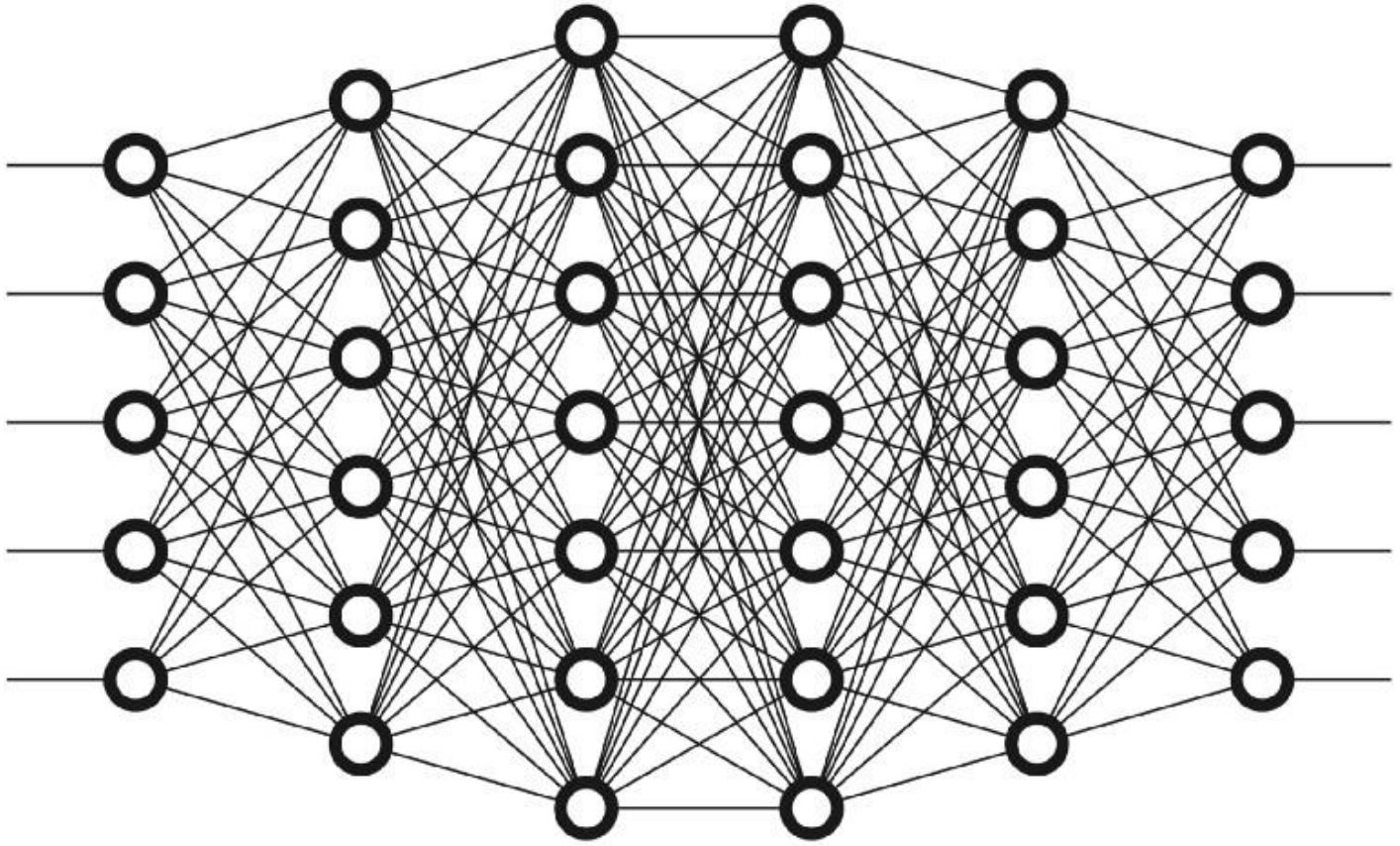
Gradient Decent on Neural Networks



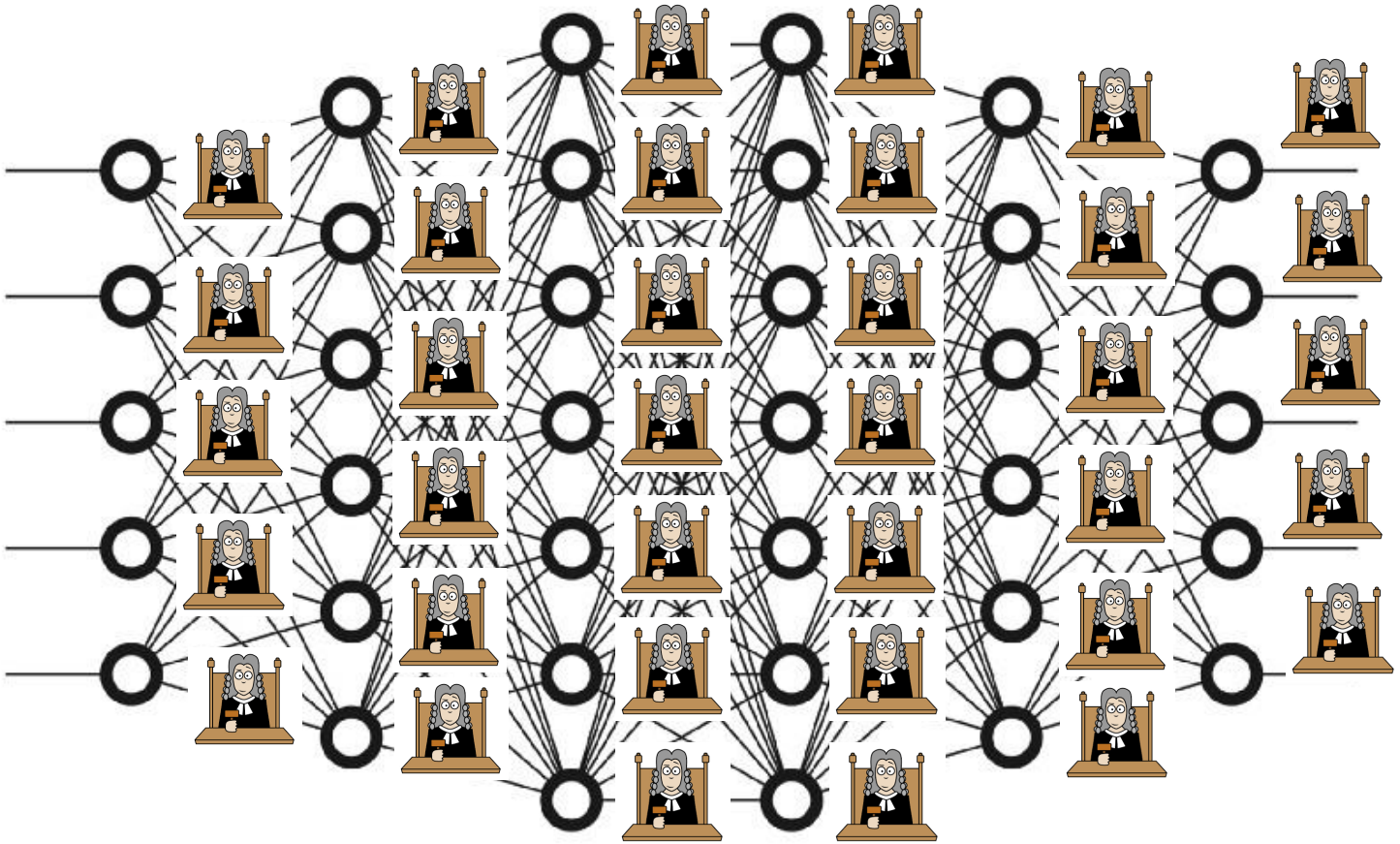
The Chain Rule for Backpropagation



By stacking more layers you have **Deep Neural Networks**



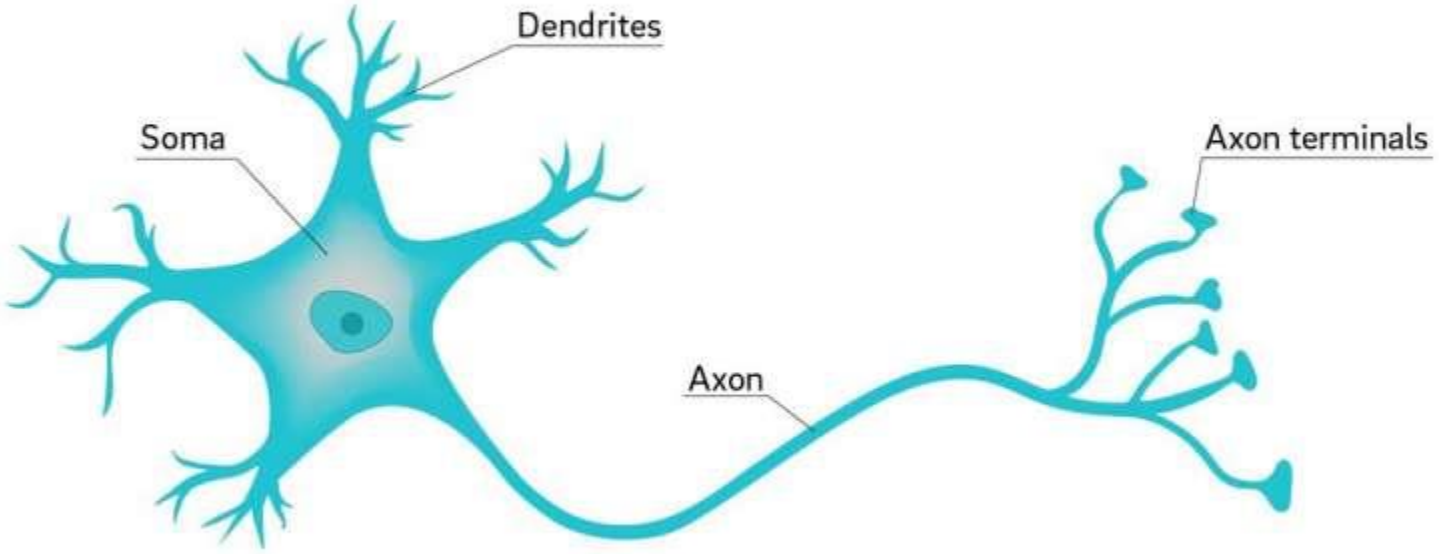
By employing more loss judges you have **Deep Learning**



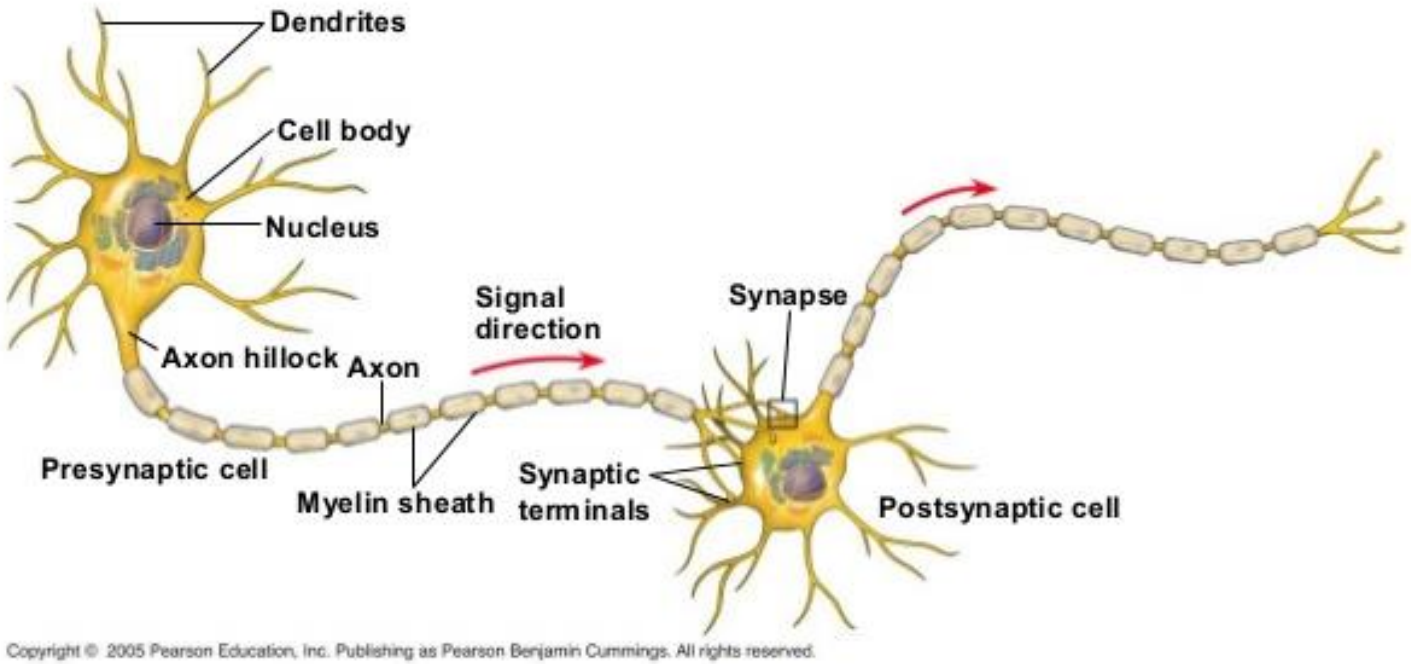


Now, we’re ready for a few more concepts (tricks)?

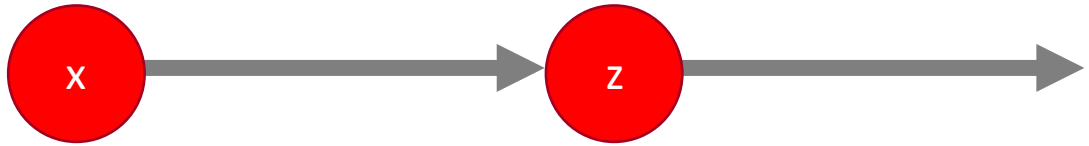
Neuron



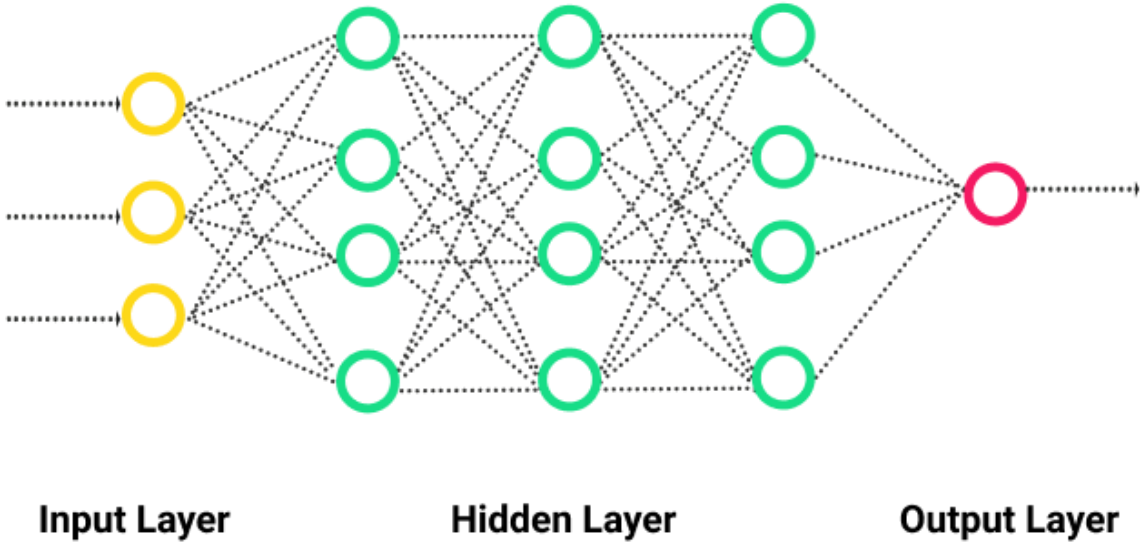




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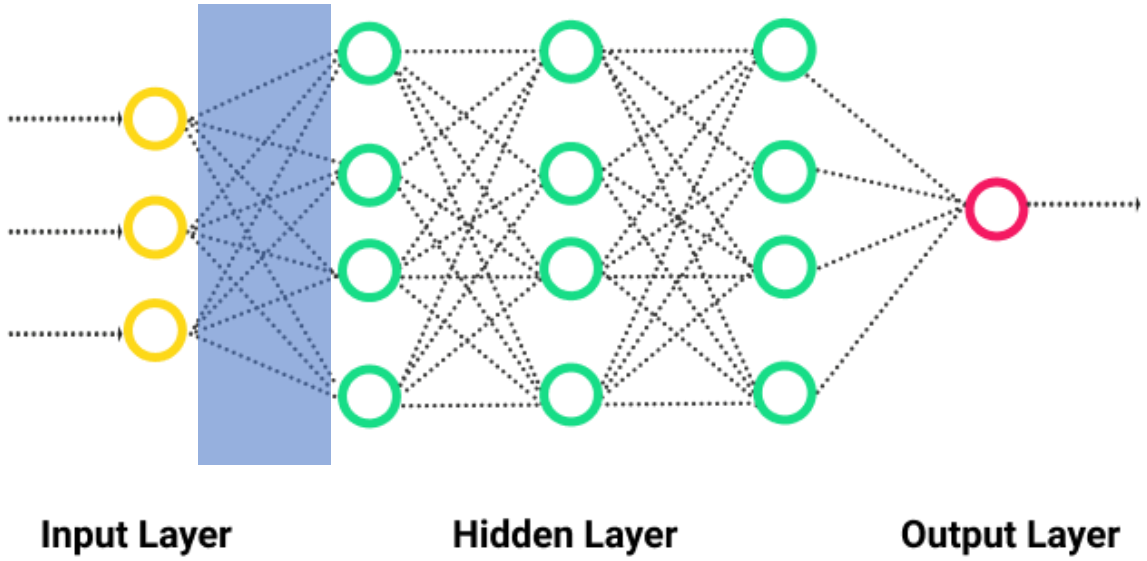
Layers



Input layer : Receives data from external sources (data files, images, sensors, etc.)
Hidden layers : Process data
Output layer: Provides network-based functions for one or more data points

https://ipullrank.com/resources/guides-ebooks/machine-learning-guide/chapter-1

Convolutional Layers



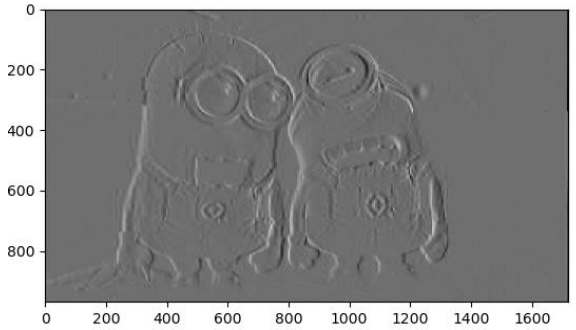
Instead of using fully connected layers, we can add a few more “partially” connected layers.

Recall the Filters and Convolutions

$$G_x = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} \qquad G_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Prewitt filter for vetical edge detection **Prewitt filter for horizontal edge detection**



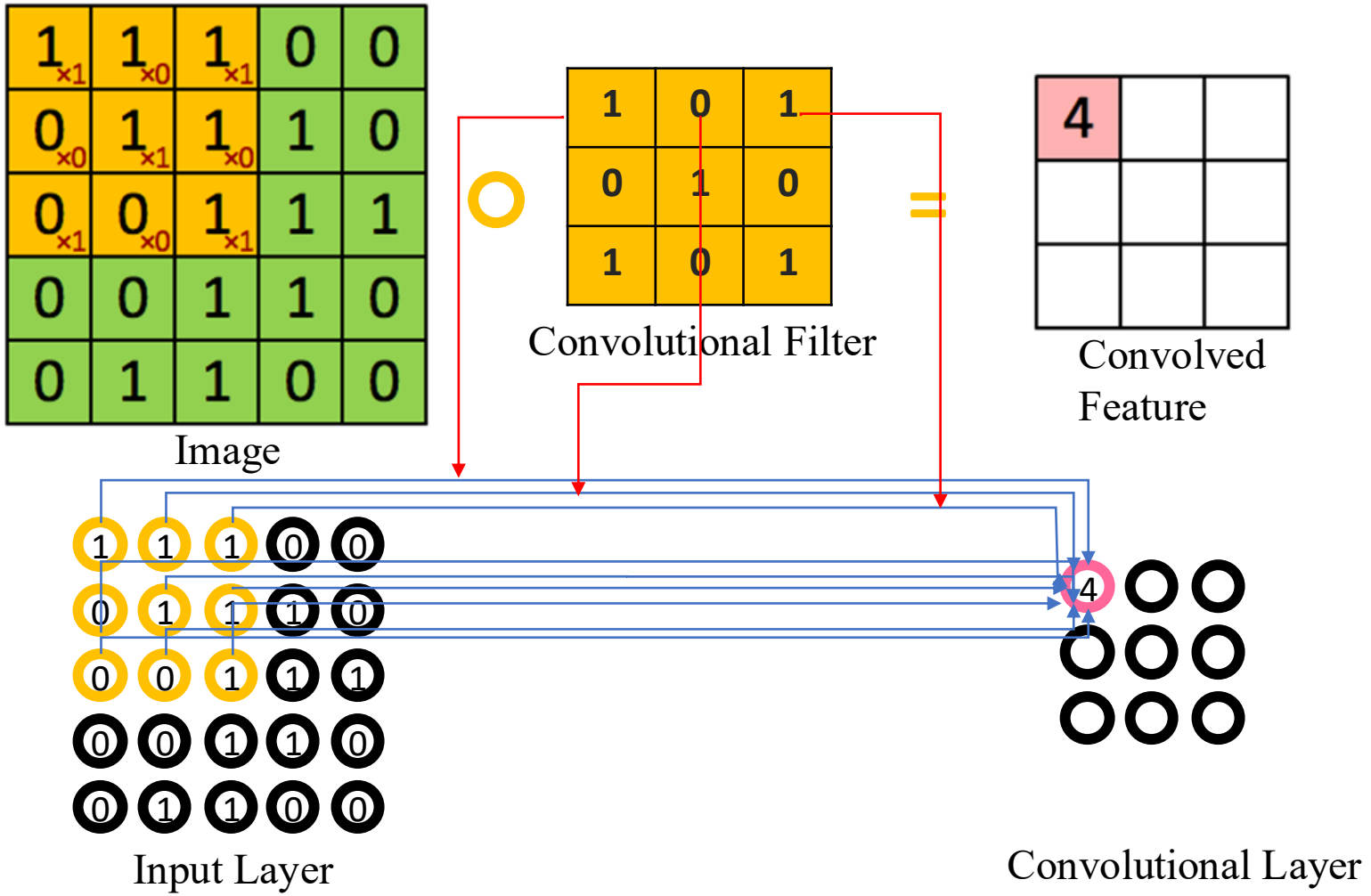


-1	0	1
-2	0	2
-1	0	1

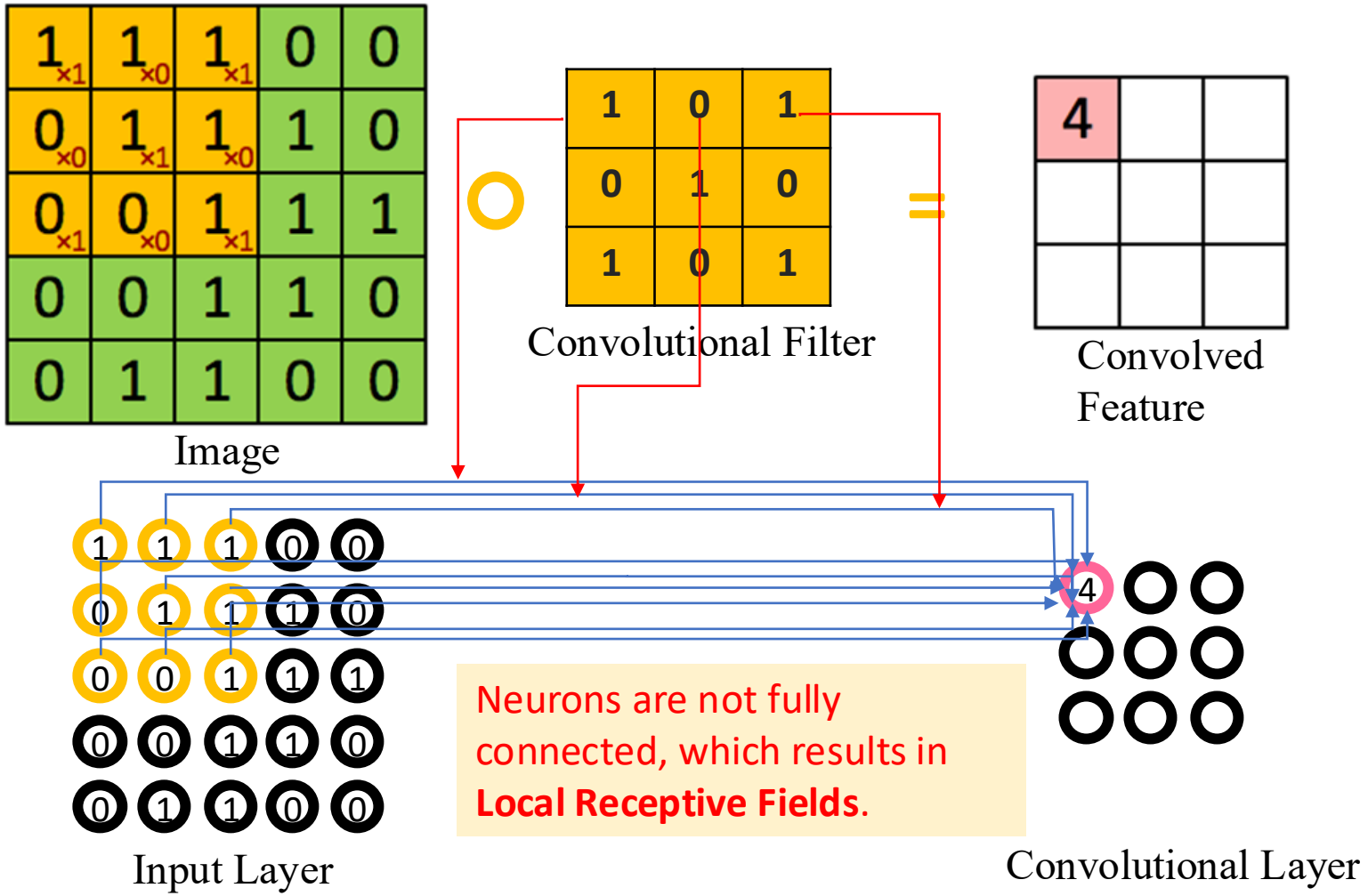
Sauber Filter



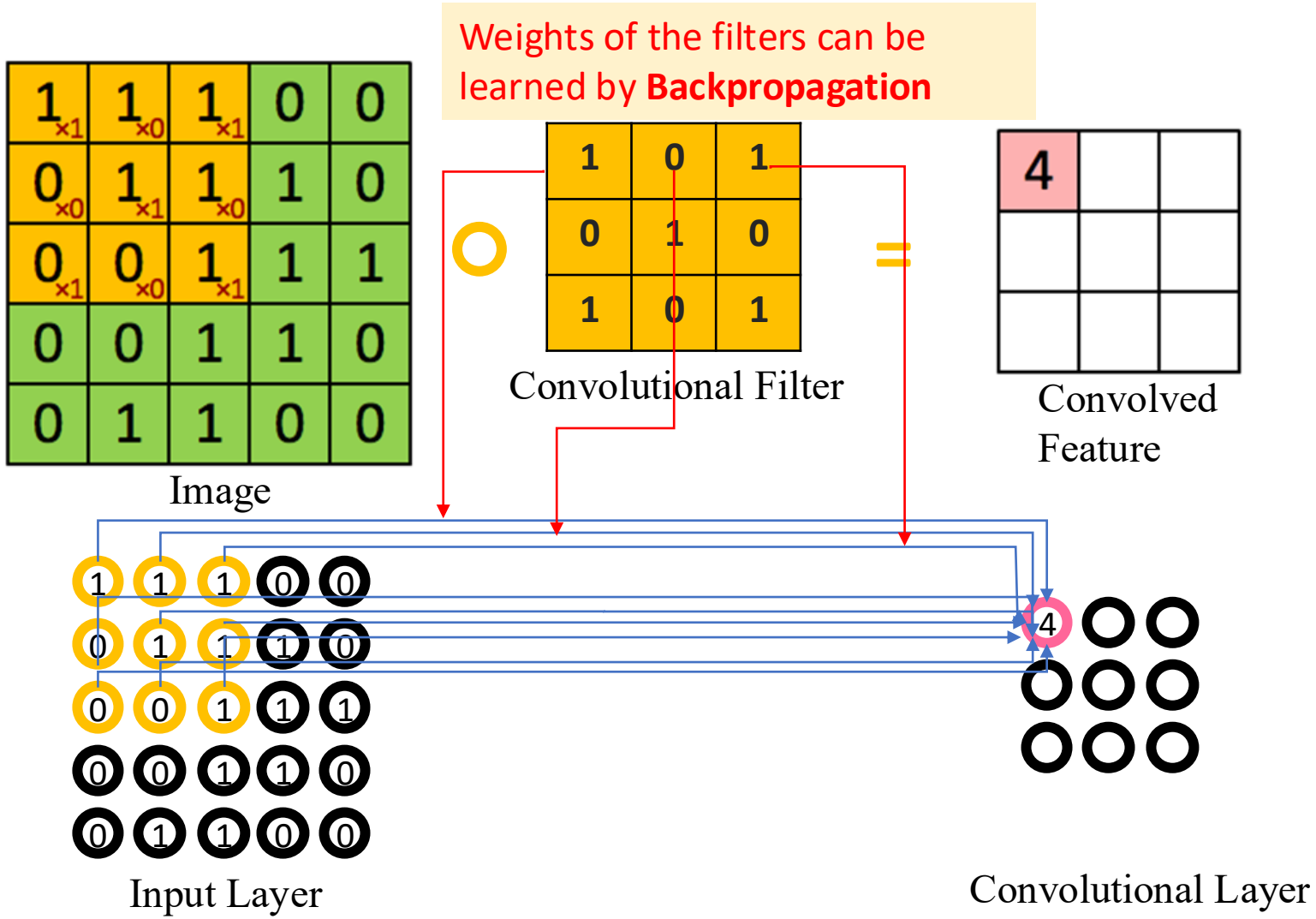
Implement with Neural Networks



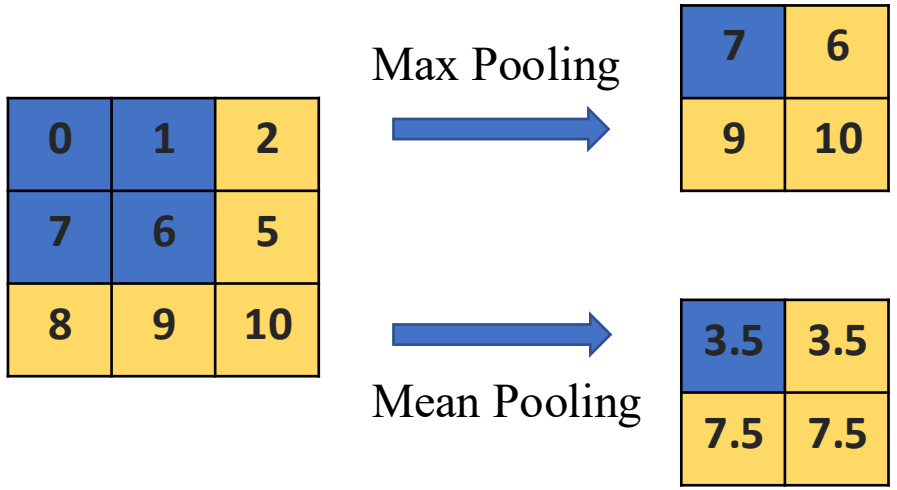
Implement with Neural Networks



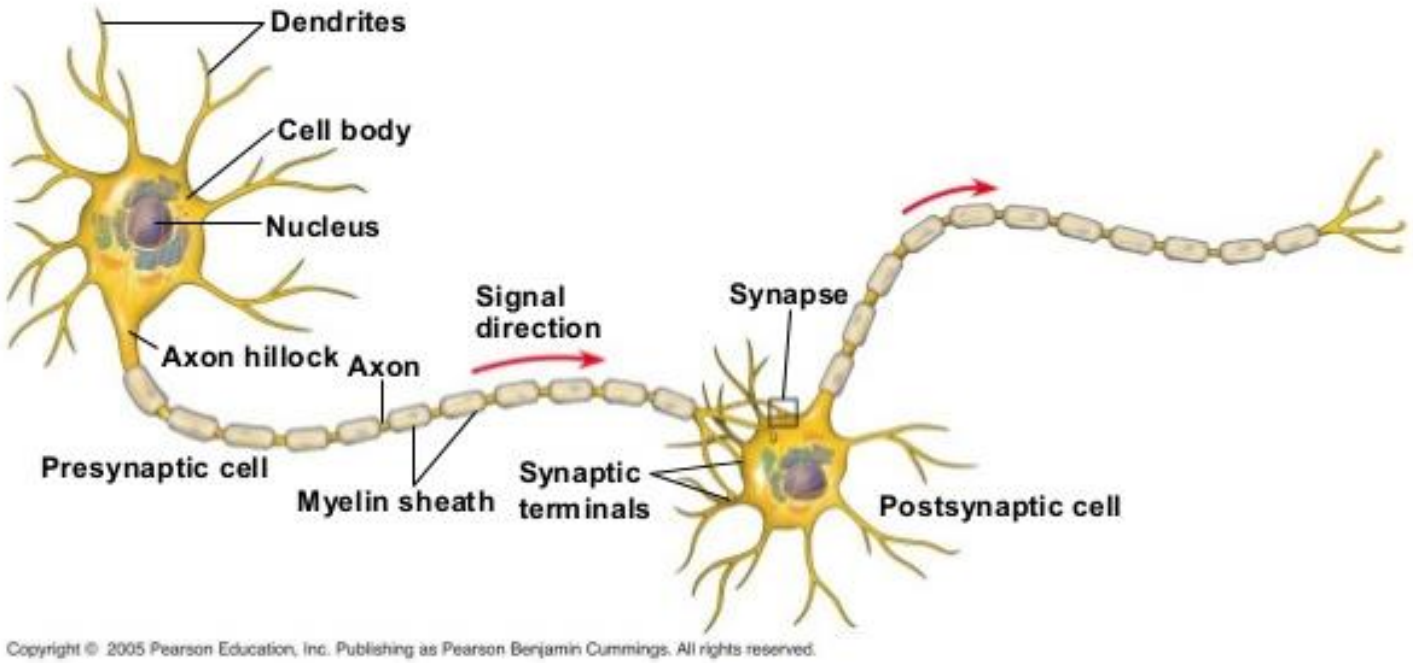
Implement with Neural Networks



Pooling

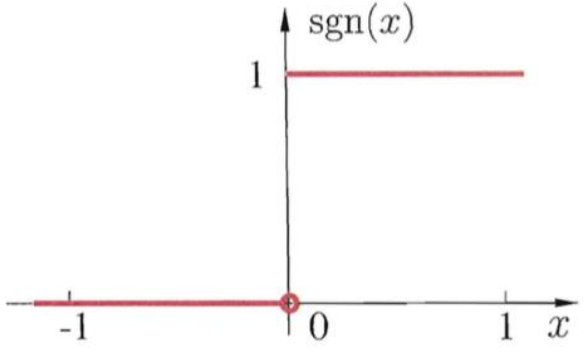


Activation

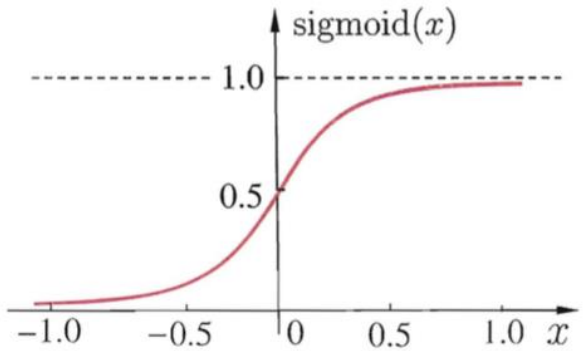


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Activation

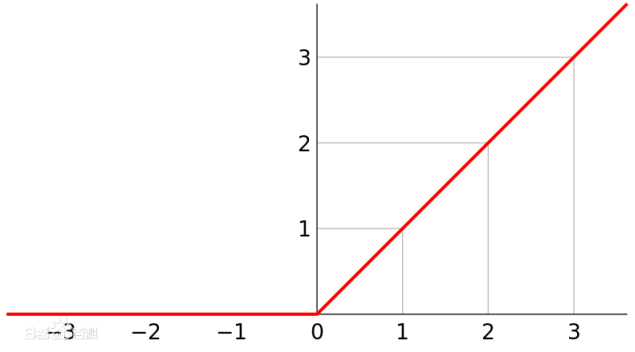


$$\operatorname{sgn}(x) = \begin{cases} 1, & x \geqslant 0; \\ 0, & x < 0. \end{cases}$$



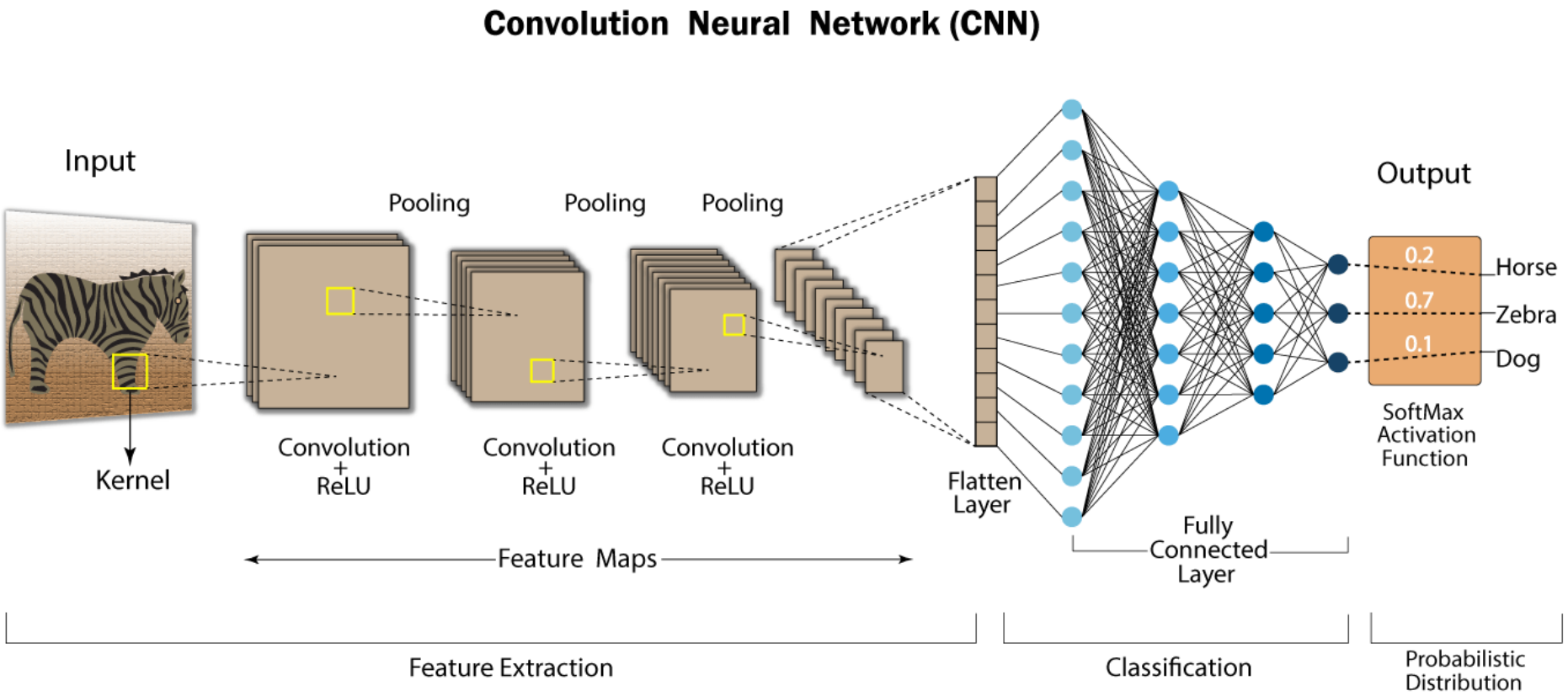
$$\operatorname{sigmoid}(x) = \frac{1}{1+e^{-x}}$$

$ReLU(x)$



$$ReLU(x) = \max(0, x)$$

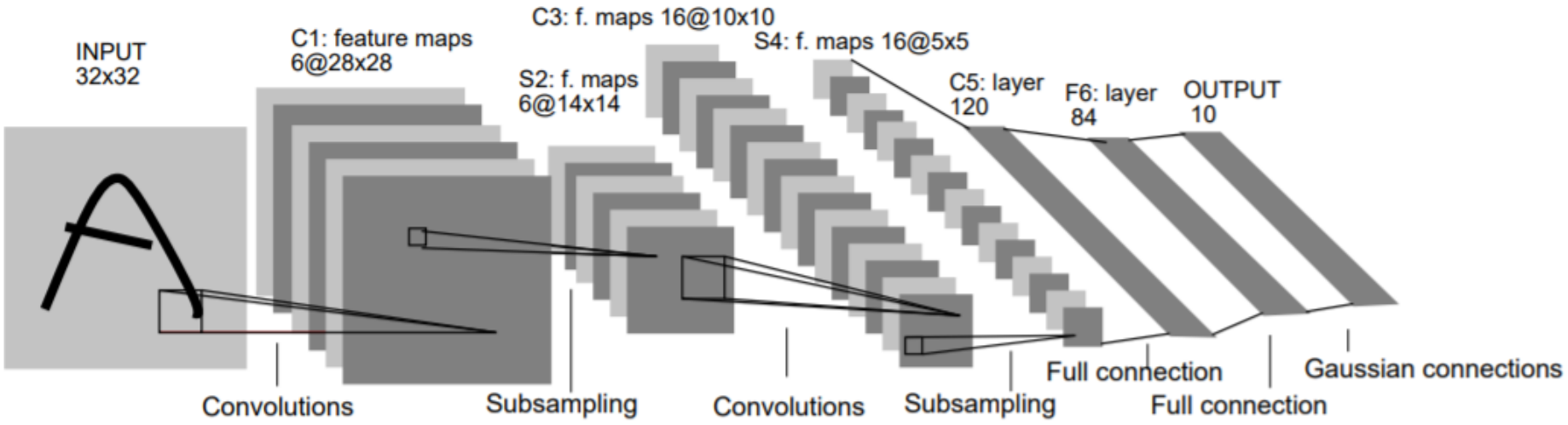
Convolutional Networks



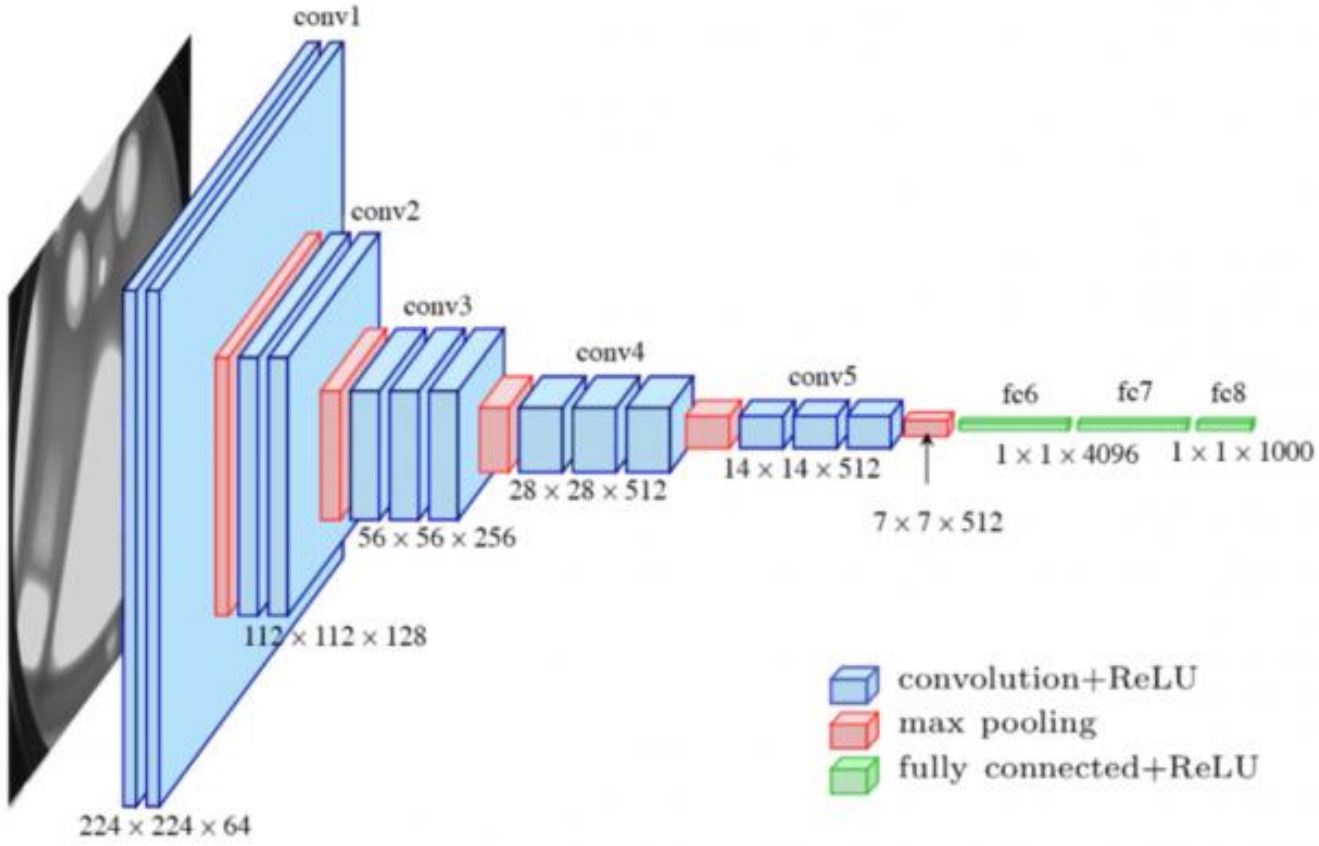
https://discuss.boardinfinity.com/t/what-do-you-mean-by-convolutional-neural-network/8533



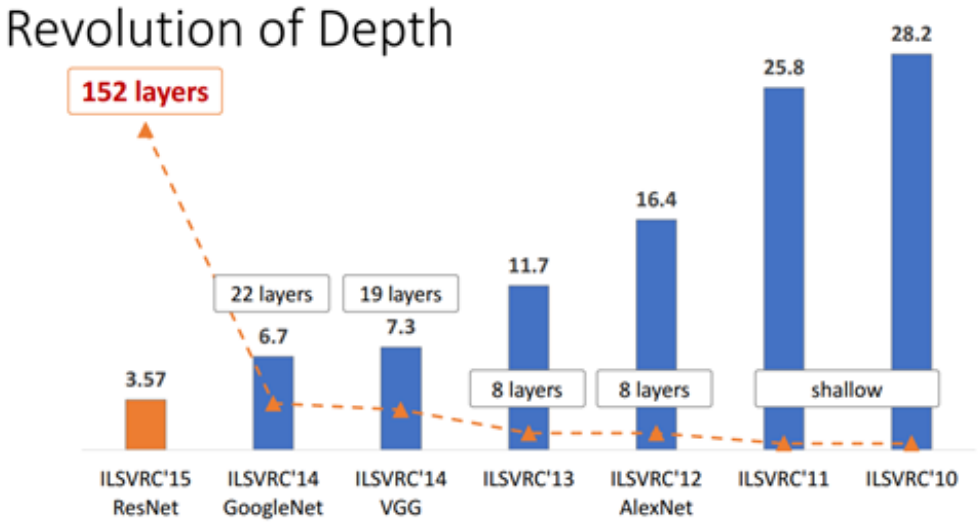
AlexNet by Alex Krizhevsky, Ilya Sutskever, and Geoff Hinton



VGG16 by Karen Simonyan, Andrew Zisserman @ Oxford



ResNet by K. He and *et al.*



ResNet @ ILSVRC & COCO 2015 Competitions

1st places in all five main tracks

ImageNet Classification: "Ultra-deep" 152-layer nets

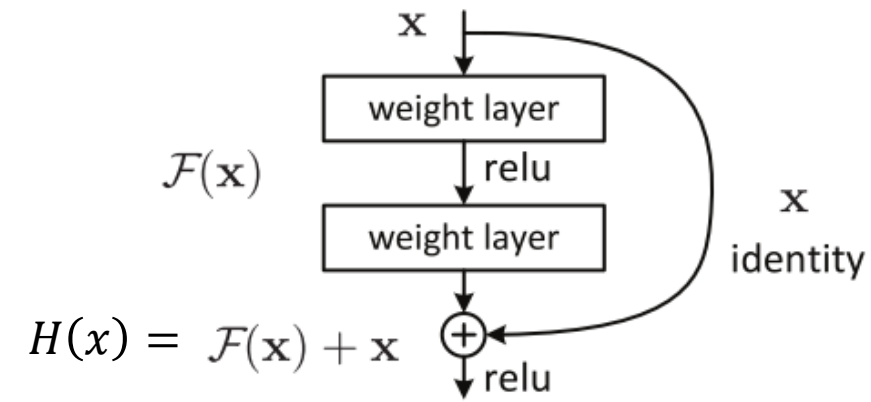
ImageNet Detection: 16% better than 2nd

ImageNet Localization: 27% better than 2nd

COCO Detection: 11% better than 2nd

COCO Segmentation: 12% better than 2nd

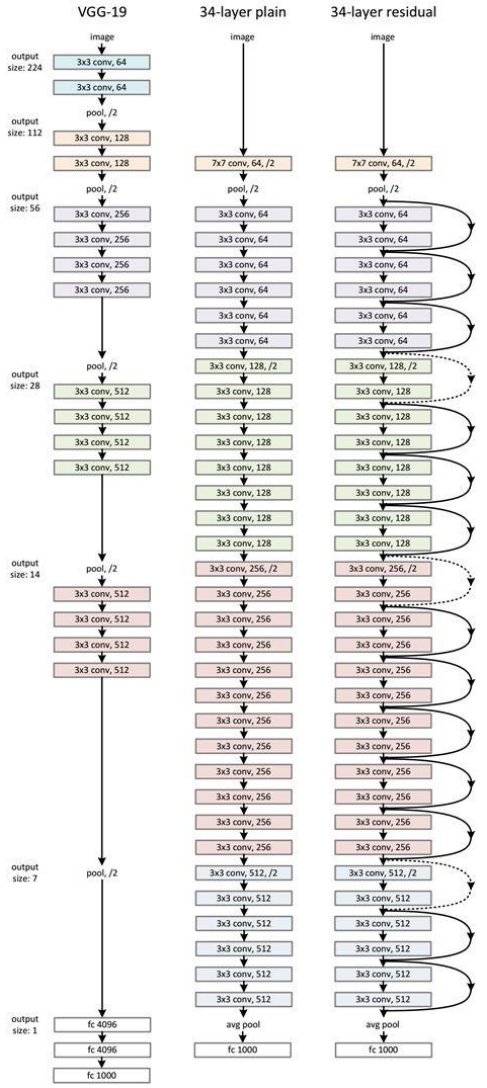
Vanishing Gradients and Residual Learning



$$H(x) = \mathcal{F}(\mathbf{x}) + \mathbf{x}$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial H(x)} \frac{\partial H(x)}{\partial x}$$

$$\frac{\partial H(x)}{\partial x} = \frac{\partial (F(x) + x)}{\partial x} = \frac{\partial F(x)}{\partial x} + 1$$



He K, Zhang X, Ren S, et al. Deep residual learning for image recognition, Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 770-778.



TODO: Study following concepts by yourself

1. Gradient vanishing and explosion
2. Adam optimizer
3. Can Gradient Decent find the optima all the time?
   Study local and local optima by yourself.

After-class Reflection and Beyond

1. **Why do traditional machine learning methods often work well on controlled datasets but struggle in real-world vision applications?**
   Consider factors such as viewpoint changes, illumination variation, occlusion, background clutter, and intra-class variation.
2. **In the red-versus-green apple example, what are the features, labels, model, and parameters?**
   Identify the roles of xxx, yyy, zzz, aaa, and bbb in the simple linear classifier.
3. **What is the purpose of a loss function in machine learning?**
   Explain why the loss function measures how “bad” a model is rather than how “good” it is.
4. **Why do we square the prediction error in the mean squared error loss?**
   Discuss what would happen if positive and negative errors were simply added together.
5. **What is gradient descent, and why is it useful for training models?**
   Describe how parameters are updated step by step to reduce the loss function.
6. **Why is the learning rate called a hyperparameter rather than a model parameter?**
   Compare the roles of a, b, and the learning rate δ.
7. **Can gradient descent always find the global optimum? Why or why not?**
   Discuss local optima, global optima, and the complexity of real loss surfaces.

After-class Reflection and Beyond

8. **Why does adding more nodes allow a neural network to represent more complex decision boundaries?**
   Explain how multiple intermediate nodes can combine several linear responses into a more flexible classifier.

9. **What is the difference between forward propagation and backpropagation?**
   Describe what happens during the forward pass and what happens during the backward pass.

10. **Why is the chain rule important for training neural networks?**
    Explain how backpropagation uses the chain rule to compute gradients for many weights.

11. **Why are activation functions necessary in neural networks?**
    Explain why stacking only linear layers is not enough to model complex nonlinear relationships.

12. **What is the main difference between a fully connected layer and a convolutional layer?**
    Discuss local receptive fields, shared weights, and why convolutional layers are suitable for images.

13. **What is the purpose of pooling in a CNN?**
    Explain how max pooling or mean pooling reduces spatial resolution and helps summarize local information.

After-class Reflection and Beyond

14. **Why is deep learning often called end-to-end learning?**
    Explain how CNNs learn both feature extraction and classification in one unified model.

15. **How do AlexNet, VGG16, and ResNet reflect the historical development of CNN architectures?**
    Compare their main contributions: large-scale deep learning, deeper networks with small filters, and residual connections.

16. **What problem does ResNet try to solve when training very deep networks?**
    Discuss vanishing gradients and why simply adding more layers may not always improve performance.

17. **How do shortcut connections in ResNet help gradient flow?**
    Explain the intuition behind

18. **If you were designing a CNN for classifying Rock-Paper-Scissors gestures, what would be learned automatically by the network compared with the landmark-distance approach from the previous lecture?**
    Compare handcrafted features and learned convolutional features.

Thank You!

